Home / Grade 11 / Algebra / Quarter 3

The definite integral and the Newton–Leibniz formula

Two limits on the integral sign, one subtraction — and the theorem that makes area computable.

LESSON 49–51 | 3 × 40 MIN | ALGEBRA 11, §3.1 | P1 · 10.4

LESSON CLOCK

18' | 26' | 26' | 26' | 30' | 9'

Area as a limit of rectangles	18 min
The Newton–Leibniz formula	26 min
Properties	26 min
Signed area	26 min
Practice	30 min
Homework	9 min

LEARNING OBJECTIVES

- Interpret the definite integral as a limit of a sum of rectangles.
- State and apply the Newton–Leibniz formula.
- Use the properties of the definite integral.
- Explain why the constant of integration cancels.

TEXTBOOK

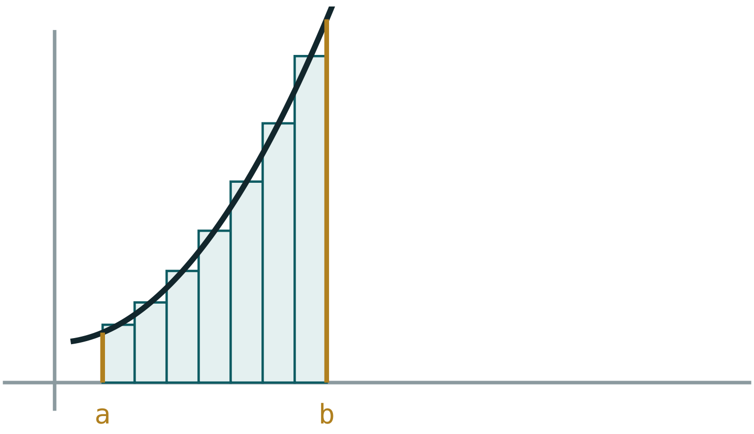
National	pp. 225–242
Cambridge	Pure Mathematics 1, pp. 217–226
Workbook	P1 Exercise 10D

01 Explanation

Area as a limit of rectangles

To find the area under $y = f(x)$ between a and b, cut the interval into n strips, replace each strip by a rectangle, and add:

$$S\_n = \Sigma f(x\_i) \cdot \Delta x$$



more, thinner rectangles → the definite integral

Seven rectangles already approximate the area. Take more and thinner, and the error vanishes.

As $n \to \infty$ and every $\Delta x \to 0$, the sum tends to a definite number — the **definite integral**:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \Sigma f(x_i)\Delta x$$

TWO DIFFERENT OBJECTS, ONE SYMBOL

The **indefinite** integral is a family of functions. The **definite** integral is a **number**. They are joined by the theorem below, and that link is the central result of the whole subject.

The Newton–Leibniz formula

THE FUNDAMENTAL THEOREM

If F is any antiderivative of f on $[a, b]$ then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

written $[F(x)]_a^b$.

Nothing about limits of sums survives in the calculation: find an antiderivative, put in the two numbers, subtract.

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3}$$

NO $+C$ IN A DEFINITE INTEGRAL

The constant cancels: $(F(b) + C) - (F(a) + C) = F(b) - F(a)$. Writing it is not wrong, but carrying it through is wasted work — and writing it in the **answer** is a mistake, because the answer is a number.

Properties

$\int_a^a f = 0$ $\int_a^b f = -\int_b^a f$

$$\int_a^b (f \pm g) = \int_a^b f \pm \int_a^b g$$

$\int_a^b f = \int_a^c f + \int_c^b f$

The last is **additivity**: an interval may be split at any interior point, and the parts add.
It is what allows a curve crossing the axis to be handled piece by piece.

Signed area

THE INTEGRAL IS A SIGNED AREA

Where the curve is **below** the axis, $f(x) < 0$ and the contribution is negative. $\int_0^{2\pi} \sin x \, dx = 0$ — not because there is no area, but because the two halves cancel.

So “find the integral” and “find the area” are different questions:

1. **Integral:** compute it directly, signs and all.
2. **Area:** find the zeros of f in $[a, b]$, integrate over each piece separately, and add the **absolute values**.

For $y = x^3$ on $[-1, 1]$: the integral is 0, but the area is $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

02 Worked examples

EXAMPLE 1 **Evaluate $\int_1^3 (2x + 1)\,dx$.**

1. $F(x) = x^2 + x$
2. $F(3) = 12$, $F(1) = 2$.
3. $12 - 2 = 10$

ANSWER

10

EXAMPLE 2 **Evaluate $\int_1^4 \frac{1}{\sqrt{x}}\,dx$.**

1. Rewrite as $x^{(-1/2)}$.
2. $F(x) = 2\sqrt{x}$
3. $2(2) - 2(1) = 2$

ANSWER

2

EXAMPLE 3 **Find the area between $y = x^3$ and the x-axis on $[-1, 1]$.**

1. The curve crosses at $x = 0$.
   Split there.
2. $\int_{-1}^{0} = -\frac{1}{4}$
3. $\int_{0}^{1} = \frac{1}{4}$
4. Add the absolute values.

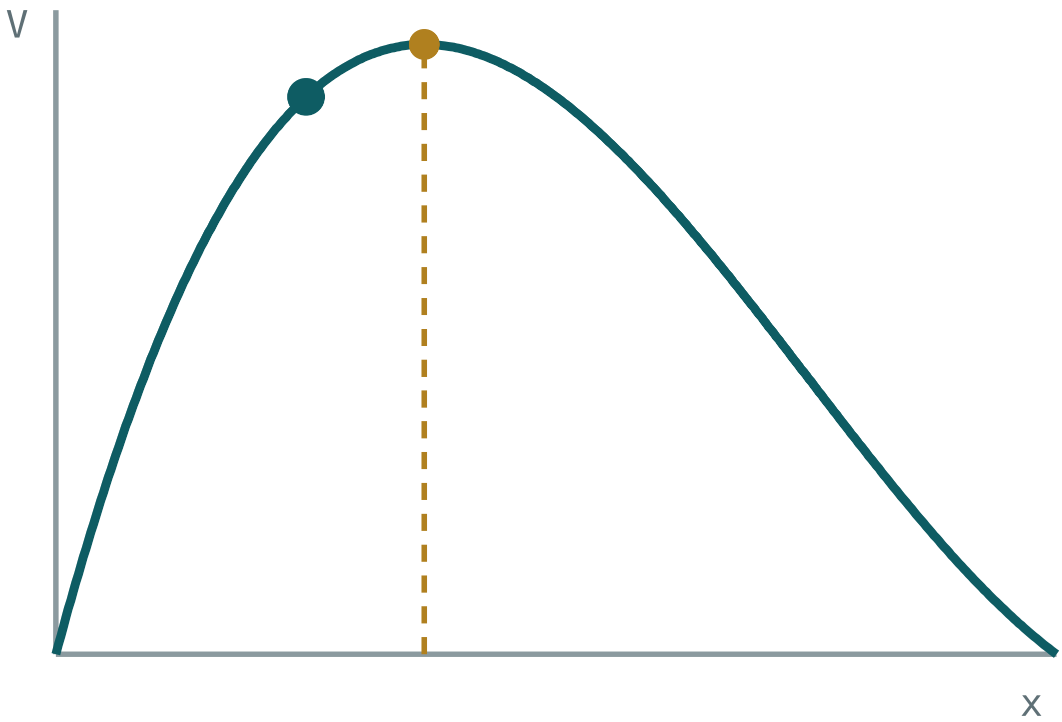
ANSWER

$\frac{1}{2}$

03 Interactive model

Increase the number of rectangles and watch the estimate settle on the exact value.

Rectangles filling an area



x **2** | V(x) **384** | greatest V **420.1** | at x = **2.94**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Definite integral	Aniq integral	Определённый интеграл
Limits of integration	Integrallash chegaralari	Пределы интегрирования
Lower limit	Quyi chegara	Нижний предел
Upper limit	Yuqori chegara	Верхний предел
Newton–Leibniz formula	Nyuton–Leybnits formulasi	Формула Ньютона–Лейбница
Integral sum	Integral yigʻindi	Интегральная сумма
Area under a curve	Egri chiziq ostidagi yuza	Площадь под кривой
Additivity	Additivlik	Аддитивность
Sign of the integral	Integral ishorasi	Знак интеграла

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A definite integral is:

- a family of functions
- a number
- a derivative
- an equation

The Newton–Leibniz formula gives:

- $F(a) - F(b)$
- $F(b) - F(a)$
- $F(b) + F(a)$
- $F'(b)$

The constant of integration in a definite integral:

- must be included
- cancels
- doubles
- is the answer

$\int_a^a f$ equals:

- $f(a)$
- 0
- $F(a)$
- undefined

Below the axis the integral contributes:

- positively
- negatively
- nothing
- twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\int_0^2 x\,dx$

ANSWER 2

2. $\int_0^3 2\,dx$

ANSWER 6

3. $\int_0^1 x^2\,dx$

ANSWER $\frac{1}{3}$

4. $\int_1^2 3x^2\,dx$

ANSWER 7

5. $\int_2^2 x^5\,dx$

ANSWER 0

6. $\int_0^4 \sqrt{x}\,dx$

ANSWER $\frac{16}{3}$

7. Does a definite integral need $+\ C$?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. $\int_1^3 (2x + 1)\, dx$

ANSWER 10

2. $\int_1^4 \frac{1}{\sqrt{x}}\, dx$

ANSWER 2

3. $\int_0^2 (x^2 - 3x)\, dx$

ANSWER $-\frac{10}{3}$

4. $\int_1^2 \frac{1}{x^2}\, dx$

ANSWER $\frac{1}{2}$

5. $\int_{-1}^1 x^3\, dx$

ANSWER 0

6. Area between $y = x^3$ and the axis on $[-1, 1]$

ANSWER $\frac{1}{2}$

7. $\int_0^1 (3x^2 + 2x)\, dx$

ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. $\int_1^2 \frac{x^2 + 1}{x^2}\, dx$

ANSWER $\frac{3}{2}$

2. $\int_0^1 (2x + 1)^3\, dx$

ANSWER 10

3. Area between $y = x^2 - 4$ and the axis on $[0, 3]$

ANSWER $\frac{23}{3}$

4. Find k with $\int_0^k x\, dx = 8$

ANSWER $k = 4$

5. Find k with $\int_1^k \frac{1}{x^2}\, dx = \frac{1}{2}$

ANSWER $k = 2$

6. Explain why $\int_0^{2\pi} \sin x\, dx = 0$

ANSWER The halves cancel

7. Show $\int_a^b f = \int_a^c f + \int_c^b f$

ANSWER Both equal $F(b) - F(a)$

07 Homework

Homework — 6 tasks

Write the square brackets with the limits on every answer.

1. Evaluate $\int_0^3 (x^2 + 2x)\, dx$.
2. Evaluate $\int_1^9 \frac{1}{\sqrt{x}}\, dx$.

3. Evaluate $\int_1^3 \frac{1}{x^2}\, dx$.
4. Find the area between $y = x^2 - 1$ and the axis on $[0, 2]$.
5. Find k with $\int_0^k 3x^2\, dx = 27$.
6. Explain in three sentences why the constant of integration cancels in a definite integral.

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Applications of the definite integral

Areas between curves, volumes of revolution, distance from velocity — one integral, four questions.

LESSON 52–54 | 3 × 40 MIN | ALGEBRA 11, §3.2 | P1 · 10.5–10.7

LESSON CLOCK

20' | 30' | 30' | 26' | 30' | 9'

Area with the axis	20 min
Area between two curves	30 min
Volumes of revolution	30 min
Motion	26 min
Practice	30 min
Homework	9 min

LEARNING OBJECTIVES

- Find the area between a curve and the x-axis, taking sign into account.
- Find the area between two curves.
- Find a volume of revolution about the x-axis.
- Recover distance from velocity and velocity from acceleration.

TEXTBOOK

National	pp. 243–260
Cambridge	Pure Mathematics 1, pp. 227–240
Workbook	P1 Exercise 10E, 10F

01 Explanation

Area with the axis

THE ROUTINE

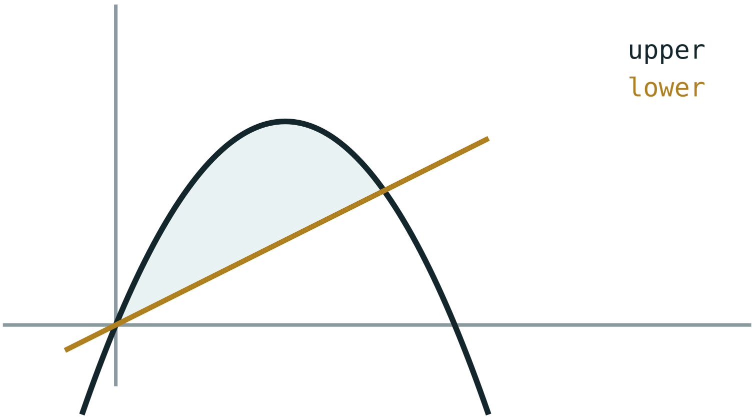
1 Find where the curve meets the axis inside the interval. **2** Integrate over each piece. **3** Add the **absolute values**.

Skipping step 1 is the standard error: a curve that dips below the axis returns a partial cancellation instead of an area.

$$\text{area} = \left|\int_a^c f\right| + \left|\int_c^b f\right| \text{ where } f(c) = 0$$

Area between two curves

$$\text{area} = \int_a^b (upper - lower)\, dx$$



Upper minus lower, integrated between the two crossing points.

The limits a and b are the x-coordinates where the curves cross, found by solving $f(x) = g(x)$ first.

Example. $y = 2x$ and $y = x^2$ meet at $x = 0$ and $x = 2$. Between them the line is above:

$$\int_0^2 (2x - x^2)\,dx = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$$

> WHY THE SIGN TAKES CARE OF ITSELF
>
> Because $upper - lower$ is positive throughout, the integral is the area directly — even if both curves are below the axis. That is why this method is safer than integrating each separately.

Volumes of revolution

Rotate the region under $y = f(x)$ between a and b about the x-axis. Each thin strip becomes a disc of radius y and thickness dx:

$$V = \pi \int_a^b y^2\,dx$$

CURVE	INTERVAL	SOLID	VOLUME
$y = r$	$[0, h]$	cylinder	$\pi r^2 h$
$y = \frac{r}{h} x$	$[0, h]$	cone	$\frac{1}{3}\pi r^2 h$
$y = \sqrt{r^2 - x^2}$	$[-r, r]$	sphere	$\frac{4}{3}\pi r^3$

Every volume formula of Grade 11 geometry is one of these integrals. The $\frac{1}{3}$ in the cone is $\int x^2\, dx$; the $\frac{4}{3}$ in the sphere is $\int (r^2 - x^2)\, dx$ over $[-r, r]$.

> SQUARE THE Y, DO NOT SQUARE THE INTEGRAL
>
> $\pi \int y^2\, dx$, not $\pi(\int y\, dx)^2$. The two differ completely.

Motion

$$s = \int v\, dt \quad v = \int a\, dt$$

Integration undoes the differentiation of Quarter I. With limits it gives a definite answer:

$$\text{displacement} = \int_{t_1}^{t_2} v\, dt \quad \text{distance} = \int_{t_1}^{t_2} |v|\, dt$$

> DISPLACEMENT IS NOT DISTANCE, AGAIN
>
> If v changes sign in the interval, split at that instant and add the absolute values — exactly as for area. It is the same theorem wearing different words.

02 Worked examples

EXAMPLE 1 **Find the area between $y = 2x$ and $y = x^2$.**

1. $2x = x^2 \Rightarrow x = 0, 2$

   The limits.

2. The line is above between them.

3. $\int_0^2 (2x - x^2)\,dx$

4. $= 4 - \frac{8}{3} = \frac{4}{3}$

ANSWER

$\frac{4}{3}$

EXAMPLE 2 **The region under $y = \sqrt{x}$ from 0 to 4 is rotated about the x-axis. Find the volume.**

1. $y^2 = x$

2. $V = \pi\int_0^4 x\,dx$

3. $= \pi\left[\frac{x^2}{2}\right]_0^4$

4. $= 8\pi$

ANSWER

$8\pi \approx 25.1$

EXAMPLE 3 **A body has $v = 3t^2 - 12t$ m/s. Find the distance travelled in the first 5 s.**

1. $v = 3t(t - 4)$, zero at $t = 0, 4$.
   Split at 4.

2. $\int_0^4 v\,dt = [t^3 - 6t^2] = -32$

3. $\int_4^5 v\,dt = (125 - 150) - (-32) = 7$

4. $32 + 7$

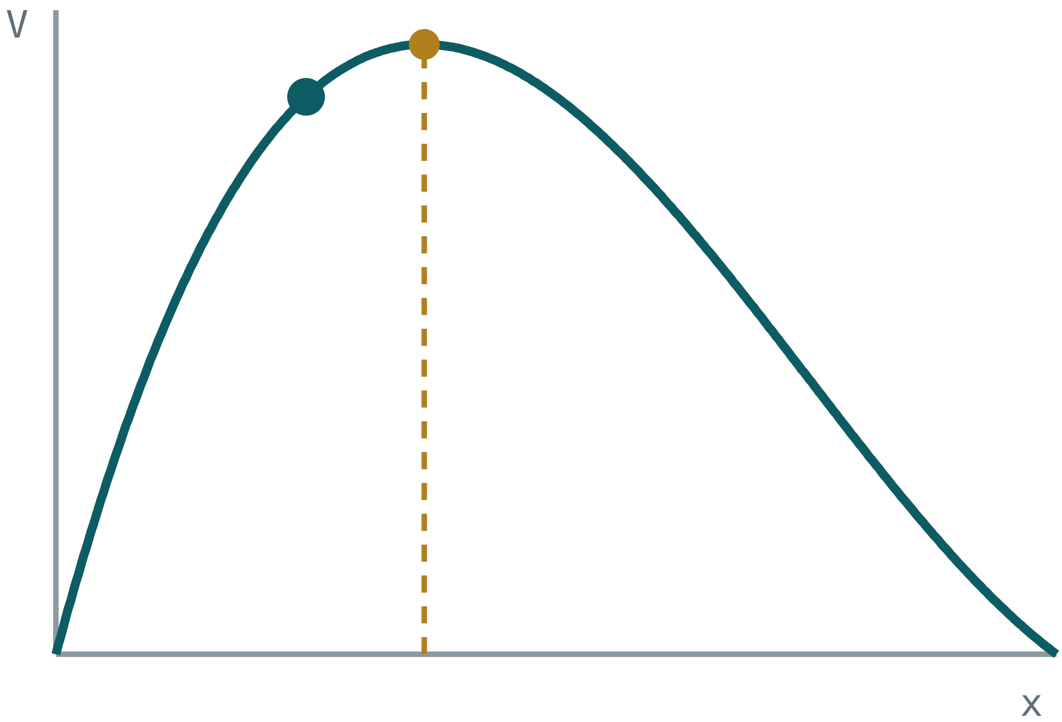
ANSWER

39 m

03 Interactive model

Sketch the two curves and shade the region before writing any integral.

Area under a curve



x **2** | V(x) **384** | greatest V **420.1** | at x = **2.94**

04 Terminology

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ENGLISH	O‘ZBEKCHA	РУССКИЙ
Area between curves	Egri chiziqlar orasidagi yuza	Площадь между кривыми
Upper and lower curve	Yuqori va quyi egri chiziq	Верхняя и нижняя кривая
Point of intersection	Kesishish nuqtasi	Точка пересечения
Volume of revolution	Aylanish jismi hajmi	Объём тела вращения
Axis of revolution	Aylanish o‘qi	Ось вращения
Distance travelled	Bosib o‘tilgan yo‘l	Пройденный путь
Displacement	Ko‘chish	Перемещение
Signed area	Ishorali yuza	Площадь со знаком
Integrand	Integral ostidagi funksiya	Подынтегральная функция

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The area between two curves is:

- $\int f + \int g$
- $\int(upper - lower)$
- $\int|f|$
- $\int f \cdot g$

The limits come from:

- the question always
- solving $f = g$
- the axis
- guessing

A volume of revolution about Ox is:

- $\pi\int y\,dx$
- $\pi\int y^2\,dx$
- $(\pi\int y\,dx)^2$
- $\int y^2\,dx$

Distance from velocity is:

- $\int v\, dt$
- $\int |v|\, dt$
- v'
- $\int v^2\, dt$

A curve dipping below the axis needs:

- nothing special
- splitting at the zeros
- squaring
- doubling

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Area under $y = x$ from 0 to 4

   ANSWER 8

2. Area under $y = x^2$ from 0 to 3

   ANSWER 9

3. Area under $y = 2$ from 1 to 5

   ANSWER 8

4. Volume from $y = 2$ on $[0, 3]$ about Ox

   ANSWER 12π

5. Volume from $y = x$ on $[0, 3]$

   ANSWER 9π

6. $v = 4$; distance in 6 s

   ANSWER 24 m

7. $v = 2t$; distance in 3 s

   ANSWER 9 m

Medium · the standard question

7 PROBLEMS

1. Area between $y = 2x$ and $y = x^2$

ANSWER $\frac{4}{3}$

2. Area between $y = x$ and $y = x^2$

ANSWER $\frac{1}{6}$

3. Area between $y = 4 - x^2$ and the axis

ANSWER $\frac{32}{3}$

4. Volume from $y = \sqrt{x}$ on $[0, 4]$

ANSWER 8π

5. Volume from $y = x^2$ on $[0, 2]$

ANSWER $\frac{32\pi}{5}$

6. $v = 3t^2 - 12t$; distance in the first 5 s

ANSWER 39 m

7. Same; the displacement

ANSWER -25 m

Hard · stretch and reason

7 PROBLEMS

1. Area between $y = x^2$ and $y = x^3$

   ANSWER $\frac{1}{12}$

2. Area between $y = x^2 - 4$ and $y = 5$

   ANSWER 36

3. Volume from $y = \frac{r}{h}x$ on $[0, h]$

   ANSWER $\frac{1}{3}\pi r^2 h$

4. Volume from $y = \sqrt{r^2 - x^2}$ on $[-r, r]$

   ANSWER $\frac{4}{3}\pi r^3$

5. Area enclosed by $y = x^3 - x$

   ANSWER $\frac{1}{2}$

6. $a = 6t$, $v(0) = 0$, $s(0) = 2$. Find $s(3)$

   ANSWER 29

7. Volume from $y = \frac{1}{x}$ on $[1, 3]$ about Ox

   ANSWER $\frac{2\pi}{3}$

07 Homework

Homework — 6 tasks

Sketch and shade before every integral; find the limits before writing them.

1. Find the area between $y = 3x$ and $y = x^2$.
2. Find the area between $y = 9 - x^2$ and the x-axis.

3. Find the volume when the region under $y = x$ on $[0, 4]$ is rotated about Ox.
4. Find the volume when the region under $y = \sqrt{4 - x^2}$ on $[-2, 2]$ is rotated about Ox, and name the solid.
5. A body has $v = t^2 - 4t$ m/s. Find the displacement and the distance in the first 6 s.
6. Explain in three sentences why “upper minus lower” needs no absolute value.

The trapezium rule

Cambridge insert: what to do when the antiderivative does not exist — and how to judge the error.

LESSON 55–57 | 3 × 40 MIN | ALGEBRA 11, §3.2 (EXTENSION) | P2 · 5.7

LESSON CLOCK

18' | 28' | 28' | 26' | 30' | 5

Why a numerical method	18 min
The rule	28 min
Over or under?	28 min
Improving the estimate	26 min
Practice	30 min
Homework	5 min

LEARNING OBJECTIVES

- State and apply the trapezium rule with n strips.
- Decide whether the estimate is an over- or an under-estimate.
- Improve an estimate by doubling the number of strips.
- Use the rule on data given only as a table.

TEXTBOOK

National	pp. 261–268
Cambridge	Pure Mathematics 2 & 3, pp. 108–116
Workbook	P2 Exercise 5E

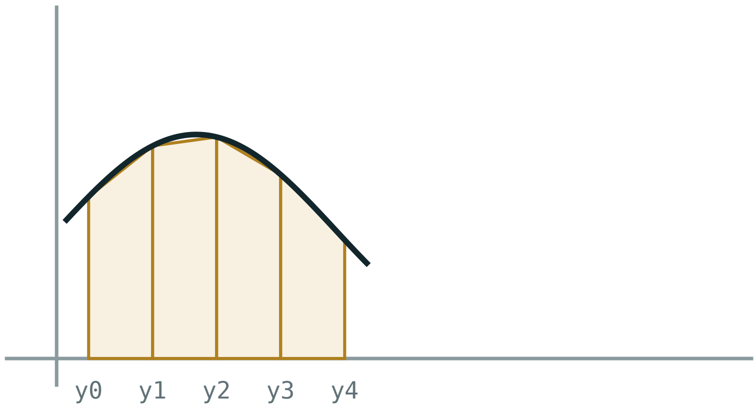
01 Explanation

Why a numerical method

Not every function has an antiderivative that can be written down. $\int \sqrt{1 + x^3}\ dx$ and $\int 2^{(x^2)}\ dx$ exist as areas but have no elementary formula. And data collected by measurement has no formula at all.

THE IDEA

Replace the curve on each strip by the **straight line** joining its ends. Each strip becomes a trapezium, whose area is elementary.



straight tops, not flat ones — the trapezium rule

Straight tops, not flat ones. Four strips already fit closely.

The rule

$$\int_a^b y\,dx \approx \frac{h}{2}\left[y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})\right]$$

with $h = \dfrac{b-a}{n}$. The **end** ordinates count once; every **interior** ordinate counts twice, because it is the right edge of one trapezium and the left edge of the next.

> N STRIPS MEANS $N+1$ ORDINATES
>
> Four strips need five values of y. Counting the strips as ordinates is the commonest arithmetic error, and it makes h wrong as well.

Example. $\int_0^4 x^2\,dx$ with 4 strips: $h = 1$, ordinates $0, 1, 4, 9, 16$.

$$\frac{1}{2}[0 + 16 + 2(1 + 4 + 9)] = \frac{1}{2}(16 + 28) = 22$$

The exact value is $\dfrac{64}{3} \approx 21.33$. The estimate is high, and the next section says why that was predictable.

Over or under?

> THE RULE OF THUMB
>
> Where the curve is **concave up** (bending upwards, $f'' > 0$), the chord lies **above** the curve, so the trapezium rule **over-estimates**. Where it is concave down, it under-estimates.

$y = x^2$ is concave up everywhere, so $22 > 21.33$ — exactly as predicted. This is a real mark in the examination: “state, with a reason, whether your estimate is an over- or an under-estimate.”

CURVE ON $[A, B]$	SHAPE	ESTIMATE
$y = x^2$	concave up	over
$y = \sqrt{x}$	concave down	under
$y = \frac{1}{x}, x > 0$	concave up	over
$y = x^3, x > 0$	concave up	over

Improving the estimate

Halving h — doubling the number of strips — divides the error by about **four**. The error is proportional to h^2.

STRIPS	H	ESTIMATE OF $\int_0^4 X^2\,DX$	ERROR
2	2	24	2.67
4	1	22	0.67
8	0.5	21.5	0.17

Each doubling quarters the error — visible in the last column. The pattern is what tells a computer when to stop.

From a table. When the data is measured rather than given by a formula, the rule applies unchanged: it needs only equally spaced ordinates, and never the formula itself.

02 Worked examples

EXAMPLE 1 **Estimate $\int_0^4 x^2\,dx$ with 4 strips.**

1. $h = 1$; ordinates $0, 1, 4, 9, 16$.

   Five of them.

2. $\frac{1}{2}[0 + 16 + 2(1 + 4 + 9)]$

3. $= \frac{1}{2}(44) = 22$

4. Concave up, so an over-estimate.

   Exact 21.33.

ANSWER

22, an over-estimate

EXAMPLE 2 **Estimate $\int_1^3 \frac{1}{x}\,dx$ with 4 strips.**

1. $h = 0.5$; ordinates $1, \frac{2}{3}, 0.5, 0.4, \frac{1}{3}$.

2. $\frac{0.5}{2}[1 + 0.3333 + 2(0.6667 + 0.5 + 0.4)]$

3. $= 0.25 \times 4.4667$

4. ≈ 1.117

   Exact $\ln 3 \approx 1.0986$.

ANSWER

≈ 1.117, an over-estimate

EXAMPLE 3 **A car's speed is measured every 10 s: $0, 8, 14, 18, 20$ m/s. Estimate the distance.**

1. $h = 10$, 4 strips, 5 ordinates.
2. $5[0 + 20 + 2(8 + 14 + 18)]$
3. $= 5(20 + 80)$
4. $= 500$ m.

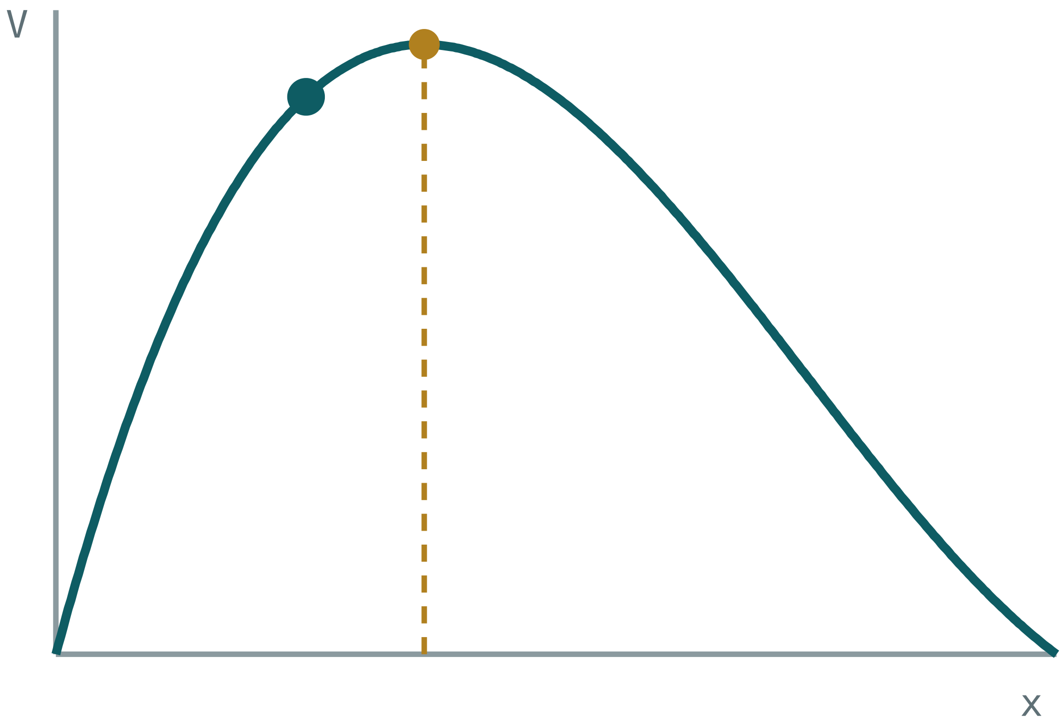
ANSWER

500 m

03 Interactive model

Compute the same integral with 2, 4 and 8 strips and put the errors side by side.

Strips under a curve



x **2** | V(x) **384** | greatest V **420.1** | at x = **2.94**

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ENGLISH	O‘ZBEKCHA	РУССКИЙ
Trapezium rule	Trapetsiya usuli	Метод трапеций
Strip	Bo‘lak	Полоса
Ordinate	Ordinata	Ордината
Strip width h	Bo‘lak kengligi	Шаг
Over-estimate	Ortiqcha baho	Оценка с избытком
Under-estimate	Kam baho	Оценка с недостатком
Concave up	Yuqoriga qavariq	Выпуклая вниз
Numerical integration	Sonli integrallash	Численное интегрирование
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The trapezium rule replaces the curve by:

- flat tops
- straight chords
- parabolas
- circles

n strips need how many ordinates?

- n
- $n - 1$
- $n + 1$
- $2n$

Interior ordinates are counted:

- once
- twice
- three times
- not at all

For a concave-up curve the estimate is:

- exact
- too large
- too small
- unpredictable

Doubling the strips divides the error by about:

- 2
- 4
- 8
- 10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 4 strips need how many ordinates?

   ANSWER 5

2. 6 strips on $[0, 3]$: find h

   ANSWER 0.5

3. 5 strips on $[1, 6]$: find h

   ANSWER 1

4. Interior ordinates are counted

   ANSWER twice

5. End ordinates are counted

   ANSWER once

6. $y = x^2$ is concave

   ANSWER up

7. So the estimate is

   ANSWER too large

Medium · the standard question

7 PROBLEMS

1. $\int_0^4 x^2\,dx$ with 4 strips

   ANSWER 22

2. Its exact value

   ANSWER $\frac{64}{3} \approx 21.33$

3. $\int_0^4 x^2\,dx$ with 2 strips

   ANSWER 24

4. $\int_1^3 \frac{1}{x}\,dx$ with 4 strips

   ANSWER ≈ 1.117

5. Speeds $0, 8, 14, 18, 20$ every 10 s: the distance

   ANSWER 500 m

6. $\int_0^1 \sqrt{1 + x^3}\,dx$ with 4 strips

   ANSWER ≈ 1.111

7. Is that an over- or under-estimate?

   ANSWER over — the curve is concave up

Hard · stretch and reason

7 PROBLEMS

1. $\int_0^2 \frac{1}{1 + x^2}\, dx$ with 4 strips

   ANSWER ≈ 1.1032

2. Its exact value

   ANSWER $\arctan 2 \approx 1.1071$

3. Why is that one an under-estimate?

   ANSWER The curve is concave down over most of the range

4. $\int_0^4 x^2\, dx$ with 8 strips, and the error

   ANSWER 21.5, error 0.17

5. Show the error is about h^2 times a constant

   ANSWER Each doubling quarters it

6. Widths $0, 2.4, 3.1, 3.6, 3.9, 4.0$ m every 5 m: the area

   ANSWER ≈ 65.5 m²

7. A curve is concave up then down. What can be said about the estimate?

   ANSWER Nothing in general — split the interval

07 Homework

Homework — 5 tasks

Always state the number of strips, the value of h, and whether the estimate is over or under.

1. Estimate $\int_0^3 x^3\, dx$ with 3 strips and compare with the exact value.
2. Estimate $\int_1^5 \frac{1}{x}\, dx$ with 4 strips, and say whether it is an over- or an under-estimate.
3. Estimate $\int_0^2 \sqrt{1 + x^3}\, dx$ with 4 strips.
4. A river's depth every 4 m across is $0, 1.8, 2.6, 3.0, 2.4, 0$ m. Estimate the cross-sectional area.

5. Explain in three sentences why doubling the number of strips divides the error by about four.

Control work 5, and work on the mistakes

The definite integral in one paper — where the limits and the sketch carry the marks.

LESSON 58–59 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 5 | P1 · CHAPTER 10 REVIEW

LESSON CLOCK

3 | 40' | 10' | 25' | 12'

Instructions	3 min
The paper	40 min
Answers	10 min
Rewrite	25 min
The limits drill	12 min

LEARNING OBJECTIVES

- Apply the definite-integral methods under time.
- Find the limits of integration from the geometry, unprompted.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 269–272
Cambridge	Pure Mathematics 1, pp. 241–244
Workbook	Control paper E

01 Explanation

The paper — 25 marks, 40 minutes

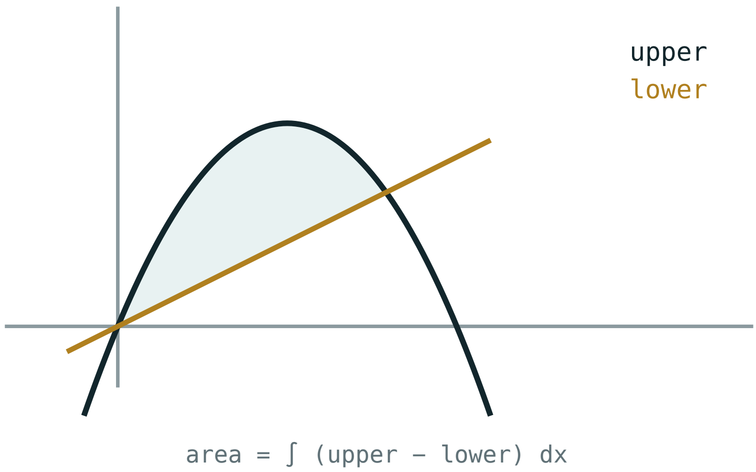
Q	TASK	MARKS	FROM
1	Evaluate $\int_1^4 (x^2 - 2x)\,dx$ and $\int_1^9 \frac{1}{\sqrt{x}}\,dx$	5	L49–51
2	Find the area between $y = 4x$ and $y = x^2$	5	L52–54
3	Find the area between $y = x^2 - 4$ and the axis on $[0, 3]$	5	L52–54
4	Volume when $y = \sqrt{x}$ on $[0, 9]$ is rotated about Ox	4	L52–54
5	Estimate $\int_0^2 \sqrt{1 + x^3}\,dx$ with 4 strips, and say over or under	6	L55–57

WHERE THE MARKS HIDE

Q2 gives one mark for solving $4x = x^2$ to get the limits. Q3 gives two for splitting at $x = 2$. Q5 gives two for the over/under statement **with its reason**. Five of twenty-five are for the setting up, not the arithmetic.

The four errors

ERROR	LOOKS LIKE	KIND
no split at a zero	Q3 answered as one integral, giving -3	method
limits guessed	Q2 done on $[0, 1]$	method
$+ C$ in a definite answer	“$= 10 + C$”	knowledge
ordinates miscounted	4 strips with 4 values of y	careless



Question 2 in one picture — and the two limits it hands you.

The limits drill

Eight pairs of curves on the board. For each, the class calls out only **the limits**, in five seconds:

CURVES	LIMITS
$y = x, y = x^2$	$0, 1$
$y = 4x, y = x^2$	$0, 4$
$y = 9 - x^2, y = 0$	$-3, 3$
$y = x^2 - 4, y = 0$	$-2, 2$
$y = x^3, y = x$	$-1, 0, 1$

THE LAST ROW HAS THREE

$x^3 = x$ at $-1, 0, 1$, and the curves swap over at 0. Two integrals are needed, or the parts will cancel.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: area between $y = x^2 - 4$ and the axis on $[0, 3]$.**

1. Zero at $x = 2$ inside the interval.

   Split. Two marks.

2. $\int_0^2 = \frac{8}{3} - 8 = -\frac{16}{3}$

3. $\int_2^3 = (9 - 12) - (-\frac{16}{3}) = \frac{7}{3}$

4. $\frac{16}{3} + \frac{7}{3} = \frac{23}{3}$

ANSWER

$\frac{23}{3} \approx 7.67$

EXAMPLE 2 **Model answer, Q2: area between $y = 4x$ and $y = x^2$.**

1. $4x = x^2 \Rightarrow x = 0, 4$

   One mark.

2. The line is above between them.

3. $\int_0^4 (4x - x^2)\,dx = [2x^2 - \frac{x^3}{3}]$

4. $= 32 - \frac{64}{3} = \frac{32}{3}$

ANSWER

$\frac{32}{3} \approx 10.67$

EXAMPLE 3 **A learner answered Q3 as −3. Name the mistake.**

1. They integrated straight through the zero.

2. The parts below and above the axis cancelled.

   A method error.

3. Correct: split at $x = 2$.

ANSWER

Method error — no split at the zero

03 Interactive model

Run the limits drill before the rewrite; it is the mark most often lost.

Limits, splits and signs

Limits for $y = 4x$ and $y = x^2$:

- $0, 1$
- $0, 4$
- $1, 4$
- $-4, 4$

The area between them:

- $\frac{4}{3}$
- $\frac{16}{3}$
- $\frac{32}{3}$
- 16

$y = x^2 - 4$ on $[0, 3]$ must be split at:

- 0
- 2
- 3
- nowhere

That area is:

- -3
- $\frac{23}{3}$
- 3
- $\frac{16}{3}$

Volume from $y = \sqrt{x}$ on $[0, 9]$:

- 9π
- $\frac{81\pi}{2}$
- 81π
- 18π

4 strips need how many ordinates?

- 3
- 4
- 5
- 8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Limits of integration	Integrallash chegaralari	Пределы интегрирования
Area between curves	Egri chiziqlar orasidagi yuza	Площадь между кривыми
Volume of revolution	Aylanish jismi hajmi	Объём тела вращения
Trapezium rule	Trapetsiya usuli	Метод трапеций
Careless error	E’tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Sketch	Chizma	Эскиз

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The limits of an area question come from:

- the question always
- solving for the intersections
- the axis
- guessing

A curve crossing the axis inside the interval needs:

- nothing
- a split at the zero
- squaring
- doubling

A definite integral’s answer includes:

- $+\,C$
- just a number
- a function
- a family

The trapezium estimate for a concave-up curve is:

- exact
- too large
- too small
- unpredictable

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\int_1^4 2x\,dx$

ANSWER 15

2. $\int_0^2 3x^2\,dx$

ANSWER 8

3. $\int_1^9 \frac{1}{\sqrt{x}}\,dx$

ANSWER 4

4. Limits for $y = 4x$, $y = x^2$

ANSWER $0, 4$

5. Split point for $y = x^2 - 4$ on $[0, 3]$

ANSWER $x = 2$

6. Volume from $y = \sqrt{x}$ on $[0, 9]$

ANSWER $\frac{81\pi}{2}$

7. 4 strips need how many ordinates?

ANSWER 5

Medium · the standard question 7 PROBLEMS

1. $\int_1^4 (x^2 - 2x)\,dx$

   ANSWER 6

2. Area between $y = 4x$ and $y = x^2$

   ANSWER $\frac{32}{3}$

3. Area between $y = x^2 - 4$ and the axis on $[0, 3]$

   ANSWER $\frac{23}{3}$

4. Estimate $\int_0^2 \sqrt{1 + x^3}\,dx$ with 4 strips

   ANSWER ≈ 3.28

5. Over or under, and why?

   ANSWER over — concave up

6. Area between $y = x$ and $y = x^3$ on $[0, 1]$

   ANSWER $\frac{1}{4}$

7. Volume from $y = 2x$ on $[0, 3]$

   ANSWER 12π

Hard · stretch and reason

7 PROBLEMS

1. Area enclosed by $y = x^3$ and $y = x$

   ANSWER $\frac{1}{2}$

2. Area between $y = x^2$ and $y = 2 - x^2$

   ANSWER $\frac{8}{3}$

3. Volume from $y = \frac{1}{x}$ on $[1, 4]$ about Ox

   ANSWER $\frac{3\pi}{4}$

4. Find k with area between $y = kx$ and $y = x^2$ equal to $\frac{9}{2}$

   ANSWER $k = 3$

5. Estimate $\int_1^3 \frac{1}{x}\, dx$ with 8 strips

   ANSWER ≈ 1.1032

6. Show the exact value is $\ln 3$

   ANSWER The antiderivative of $\frac{1}{x}$

7. Explain why splitting matters in Q3

   ANSWER Otherwise the two signed parts cancel

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. Find the limits before writing any integral.

1. Rewrite in full every question that lost a mark, showing how the limits were found.
2. Five problems from the section your knowledge column was heaviest in.

3. Find the area between $y = 6x$ and $y = x^2$, and between $y = x^2 - 9$ and the axis on $[0, 4]$.
4. Estimate $\int_0^1 \sqrt{1 + x^4}\, dx$ with 4 strips and state whether it is an over- or an under-estimate.

Problems in combinatorics

Counting without listing — the two rules, and the three formulas that come from them.

LESSON 60–62 | 3 × 40 MIN | ALGEBRA 11, §4.1 | EXTENSION (P&S 1)

LESSON CLOCK

The two rules	18 min
Permutations	26 min
Combinations	30 min
Does order matter?	28 min
Practice	30 min
Homework	8 min

LEARNING OBJECTIVES

- Apply the addition and multiplication rules of counting.
- Compute permutations, with and without repetition.
- Compute combinations and use the symmetry property.
- Decide whether order matters in a given problem.

TEXTBOOK

National	pp. 273–290
Cambridge	Probability & Statistics 1, pp. 82–96
Workbook	Combinatorics sheet C1

01 Explanation

The two rules

MULTIPLICATION (“AND”)

If one choice can be made in m ways and, independently, another in n ways, the pair can be made in $m \cdot n$ ways.

ADDITION (“OR”)

If a choice can be made in m ways **or**, in a mutually exclusive way, in n ways, there are $m + n$ ways in total.

Every formula below is these two rules applied repeatedly. Reading the word “and” or “or” in the problem is most of the work.

Example. A menu has 4 starters, 6 mains and 3 desserts. A full meal: $4 \times 6 \times 3 = 72$. Just one course: $4 + 6 + 3 = 13$.

Permutations

$$n! = n(n-1)(n-2)\ldots 2 \cdot 1 \quad 0! = 1$$

QUESTION	FORMULA	EXAMPLE
arrange all n in a row	$n!$	5 books: *120*
arrange k out of n, order matters	$A(n,k) = \frac{n!}{(n-k)!}$	3 of 8 runners: *336*
k choices from n, repetition allowed	n^k	4-digit PIN: 10^4
arrange n with repeats $n_1, n_2, \ldots$	$\frac{n!}{n_1!\, n_2!\, \ldots}$	MATEMATIKA

$0! = 1$ IS A DEFINITION, AND A NECESSARY ONE

It makes $A(n, n) = \frac{n!}{0!} = n!$ come out right. There is exactly one way to arrange nothing.

Combinations

$$C(n, k) = \frac{n!}{k!(n-k)!}$$

Choosing k from n when **order does not matter**. It is $A(n,k)$ divided by $k!$, because each unordered choice was counted once for every ordering.

PROPERTY	STATEMENT
symmetry	$C(n, k) = C(n, n-k)$
ends	$C(n, 0) = C(n, n) = 1$
Pascal	$C(n, k) = C(n-1, k-1) + C(n-1, k)$
total	$\Sigma C(n, k) = 2^n$

Symmetry is a labour-saving device: $C(20, 18) = C(20, 2) = 190$, computed in one line instead of ten.

Does order matter?

THE ONE QUESTION TO ASK

Would swapping two of the chosen things give a **different** outcome? If yes, order matters — use $A(n,k)$. If no, use $C(n,k)$.

SITUATION	ORDER?	FORMULA
gold, silver, bronze from 8	yes	$A(8,3) = 336$
a team of 3 from 8	no	$C(8,3) = 56$
a 4-digit code	yes, repeats allowed	10^4
a hand of 5 cards from 36	no	$C(36,5)$
seating 6 people in a row	yes	$6! = 720$

02 Worked examples

EXAMPLE 1 **From 8 runners, in how many ways can the first three places be filled?**

1. Order matters — gold is not silver.
2. $8 \times 7 \times 6$
3. $= 336$

ANSWER

336

EXAMPLE 2 **How many teams of 3 can be chosen from 8 people?**

1. Order does not matter.

2. $C(8,3) = \dfrac{8 \times 7 \times 6}{3 \times 2 \times 1}$

3. $= 56$

ANSWER

56

EXAMPLE 3 **How many arrangements has the word MATEMATIKA?**

1. 10 letters: A×3, M×2, T×2, E, I, K.

2. $\dfrac{10!}{3! \cdot 2! \cdot 2!}$

3. $= \dfrac{3\,628\,800}{24} = 151\,200$

ANSWER

$151\,200$

03 Interactive model

Ask “does swapping two of them change the answer?” before every formula.

Order or not?

Gold, silver and bronze from 8:

- $C(8,3)$
- $A(8,3)$
- 8^3
- $3!$

A team of 3 from 8:

- $C(8,3)$
- $A(8,3)$
- 8^3
- $3!$

$C(8,3)$ equals:

- 24
- 56
- 336
- 112

C(20,18) equals:

- 190
- 380
- 20
- 18

A 4-digit PIN with repeats:

- 10^4
- $A(10,4)$
- $C(10,4)$
- $4!$

0! equals:

- 0
- 1
- undefined
- ∞

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Combinatorics	Kombinatorika	Комбинаторика
Multiplication rule	Koʻpaytirish qoidasi	Правило умножения
Addition rule	Qoʻshish qoidasi	Правило сложения
Factorial	Faktorial	Факториал
Permutation	Oʻrin almashtirish	Перестановка
Arrangement	Joylashtirish	Размещение
Combination	Kombinatsiya	Сочетание
With repetition	Takrorlanish bilan	С повторениями
Order matters	Tartib muhim	Порядок важен

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The multiplication rule is used with:

- “or”
- “and”
- “not”
- neither

$C(n, k)$ equals:

- $\frac{n!}{(n-k)!}$
- $\frac{n!}{k!(n-k)!}$
- n^k
- $n!$

$C(n, k) = C(n, n-k)$ because:

- it is a coincidence
- choosing what to leave out is the same choice
- of Pascal
- of factorials

Seating 6 in a row:

- 6
- 36
- 720
- 64

$\Sigma C(n,k)$ over all k is:

- n
- $n!$
- 2^n
- n^2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $5!$

ANSWER 120

2. $0!$

ANSWER 1

3. $C(5,2)$

ANSWER 10

4. $C(6,3)$

ANSWER 20

5. $A(5,2)$

ANSWER 20

6. Arrangements of 4 books

ANSWER 24

7. 3-digit codes from 10 digits, repeats allowed

ANSWER 1000

Medium · the standard question

7 PROBLEMS

1. First three of 8 runners

   ANSWER *336*

2. Teams of 3 from 8

   ANSWER *56*

3. $C(20,18)$

   ANSWER *190*

4. Arrangements of MATEMATIKA

   ANSWER *151 200*

5. A menu: 4 starters, 6 mains, 3 desserts — full meals

   ANSWER *72*

6. Committees of 4 from 10

   ANSWER *210*

7. Arrangements of 6 people with 2 fixed together

   ANSWER *240*

Hard · stretch and reason

7 PROBLEMS

1. Hands of 5 from 36 cards

   ANSWER $C(36,5) = 376\,992$

2. Arrangements of the 8 letters of ASSALOMU with the two S apart

   ANSWER $\frac{8!}{2!} - 7! = 15\,120$

3. From 6 men and 4 women, committees of 4 with at least 1 woman

   ANSWER 195

4. How many 4-digit numbers have all digits different?

   ANSWER 4536

5. How many diagonals has a convex n-gon?

   ANSWER $\frac{n(n-3)}{2}$

6. Prove $C(n,k) = C(n-1,k-1) + C(n-1,k)$

   ANSWER Split on whether a fixed element is chosen

7. Prove $\Sigma C(n,k) = 2^n$

   ANSWER Each element is in or out

07 Homework

Homework — 6 tasks

Write “order matters” or “order does not matter” before every formula.

1. Evaluate $7!$, $C(9,4)$, $A(9,4)$ and $C(15,13)$.
2. From 10 players, how many ways to pick a team of 5? And to pick a captain, vice-captain and three others?
3. How many arrangements has the word TASHKENT? And PARALLEL?
4. How many 5-digit numbers have all digits different and do not start with 0?

5. From 7 men and 5 women, how many committees of 4 contain at least 2 women?
6. Explain in three sentences how to decide whether order matters.

The binomial theorem

Pascal's triangle, the general term, and expanding a bracket to any power without multiplying it out.

LESSON 63–64 | 2 × 40 MIN | ALGEBRA 11, §4.2 | P1 · 6.1–6.4

LESSON CLOCK

14' | 22' | 24' | 20' | 20' | 10'

Pascal's triangle	14 min
The theorem	22 min
The general term	24 min
Approximation	20 min
Practice	20 min
Homework	10 min

LEARNING OBJECTIVES

- Generate Pascal's triangle and relate it to combinations.
- Expand $(a + b)^n$ for a positive integer n.
- Find a specified term of an expansion without expanding it all.
- Use the expansion to approximate a power.

TEXTBOOK

National	pp. 291–302
Cambridge	Pure Mathematics 1, pp. 111–121
Workbook	P1 Exercise 6A–6C

01 Explanation

Pascal's triangle

Multiply out $(a + b)^n$ for small n and the coefficients form a pattern:

N	COEFFICIENTS	EXPANSION
0	1	1
1	$1\ 1$	$a + b$
2	$1\ 2\ 1$	$a^2 + 2ab + b^2$
3	$1\ 3\ 3\ 1$	$a^3 + 3a^2b + 3ab^2 + b^3$
4	$1\ 4\ 6\ 4\ 1$	$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$
5	$1\ 5\ 10\ 10\ 5\ 1$	…

EACH ENTRY IS THE SUM OF THE TWO ABOVE IT

Which is exactly $C(n,k) = C(n-1,k-1) + C(n-1,k)$ from the last lesson. The triangle **is** the table of combinations, and the coefficients are $C(n, k)$.

The theorem

$$(a + b)^n = \Sigma C(n,k)\, a^{(n-k)}\, b^k$$

Reading it term by term: the powers of a descend from n to 0, the powers of b ascend from 0 to n, every term has total degree n, and there are $n + 1$ terms.

$$(2x + 3)^3 = 8x^3 + 3(4x^2)(3) + 3(2x)(9) + 27 = 8x^3 + 36x^2 + 54x + 27$$

THE WHOLE TERM IS RAISED TO THE POWER

In $(2x + 3)^3$ the first term is $(2x)^3 = 8x^3$, not $2x^3$. Bracket the whole of a and the whole of b before applying the coefficients.

The general term

the term in b^k is $C(n,k)\, a^{(n-k)}\, b^k$

This is what makes the theorem useful: a single term can be found without expanding the rest.

Example. Find the term in x^5 in $(2x - 3)^8$.

Here $a = 2x$, $b = -3$, $n = 8$. The power of x is $8 - k$, so $k = 3$:

$$C(8,3)(2x)^5(-3)^3 = 56 \times 32x^5 \times (-27) = -48\,384\, x^5$$

The term independent of x is found the same way: write the general term, collect the powers of x, and set the total power to zero.

WATCH THE SIGNS

With $(a - b)^n$ take b to be the whole of $-b$. Odd values of k then give negative terms, and the expansion alternates.

Approximation

When x is small, the first few terms of $(1 + x)^n$ approximate it well:

$$(1 + x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$$

Example. $1.02^8 = (1 + 0.02)^8 \approx 1 + 8(0.02) + 28(0.0004) = 1 + 0.16 + 0.0112 = 1.1712$. The true value is 1.17166.

This is the linear approximation of Quarter II with one more term — and each further term improves the accuracy by roughly a factor of x.

02 Worked examples

EXAMPLE 1 **Expand $(x + 2)^4$.**

1. Coefficients $1, 4, 6, 4, 1$.
2. $x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16$
3. $= x^4 + 8x^3 + 24x^2 + 32x + 16$

ANSWER

$x^4 + 8x^3 + 24x^2 + 32x + 16$

EXAMPLE 2 **Find the term in x^5 in $(2x - 3)^8$.**

1. $k = 3$ gives $x^{(8-3)} = x^5$.
2. $C(8,3) = 56$
3. $56\,(2x)^5\,(-3)^3$
4. $= 56 \times 32 \times (-27)\,x^5 = -48\,384x^5$

ANSWER

$-48\,384x^5$

EXAMPLE 3 **Find the term independent of x in $(x^2 + \frac{1}{x})^6$.**

1. General term $C(6,k)(x^2)^{(6-k)}(x^{-1})^k$.
2. Power of x: $12 - 2k - k = 12 - 3k$.
3. $12 - 3k = 0 \Rightarrow k = 4$
4. $C(6,4) = 15$

ANSWER

15

03 Interactive model

Build Pascal’s triangle to row 6 on the board and leave it there.

Finding one term

The coefficients for $n = 4$ are:

- $1\ 4\ 4\ 1$
- $1\ 4\ 6\ 4\ 1$
- $1\ 3\ 3\ 1$
- $1\ 5\ 10\ 10\ 5\ 1$

$(x + 2)^4$ has constant term:

- 2
- 8
- 16
- 4

How many terms has $(a + b)^7$?

- 7
- 8
- 14
- 128

In $(2x + 3)^3$ the first term is:

- $2x^3$
- $8x^3$
- $6x^3$
- x^3

The term in x^5 of $(2x - 3)^8$ uses:

- $k = 5$
- $k = 3$
- $k = 8$
- $k = 0$

1.02^8 to three terms:

- 1.16
- 1.1712
- 1.18
- 1.02

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Binomial	Ikkihad	Двучлен
Binomial theorem	Nyuton binomi	Бином Ньютона
Pascal’s triangle	Paskal uchburchagi	Треугольник Паскаля
Binomial coefficient	Binomial koeffitsient	Биномиальный коэффициент
General term	Umumiy had	Общий член
Expansion	Yoyilma	Разложение
Term independent of x	x ga bog‘liq bo‘lmagan had	Член, не содержащий x
Ascending powers	Darajalar o‘sishi bo‘yicha	По возрастающим степеням
Approximation	Taqribiy hisoblash	Приближение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The binomial coefficients are:

- $n!$
- $C(n,k)$
- $A(n,k)$
- n^k

$(a + b)^n$ has how many terms?

- n
- $n + 1$
- $2n$
- 2^n

Every term has total degree:

- k
- $n - k$
- n
- $2n$

The sum of the coefficients in $(a+b)^n$ is:

- n
- $n!$
- 2^n
- n^2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Row 4 of Pascal's triangle

   ANSWER $1\ 4\ 6\ 4\ 1$

2. Row 5

   ANSWER $1\ 5\ 10\ 10\ 5\ 1$

3. Expand $(x + 1)^3$

   ANSWER $x^3 + 3x^2 + 3x + 1$

4. Expand $(x - 1)^3$

   ANSWER $x^3 - 3x^2 + 3x - 1$

5. Terms in $(a + b)^6$

   ANSWER 7

6. Constant term of $(x + 2)^4$

   ANSWER 16

7. Sum of the coefficients of $(a+b)^5$

   ANSWER 32

Medium · the standard question

7 PROBLEMS

1. Expand $(x + 2)^4$

ANSWER $x^4 + 8x^3 + 24x^2 + 32x + 16$

2. Expand $(2x - 1)^3$

ANSWER $8x^3 - 12x^2 + 6x - 1$

3. Term in x^3 of $(x + 3)^5$

ANSWER $270x^3$

4. Term in x^2 of $(2x - 1)^6$

ANSWER $60x^2$

5. Coefficient of x^4 in $(1 + x)^9$

ANSWER 126

6. 1.02^8 to three terms

ANSWER 1.1712

7. 0.98^6 to three terms

ANSWER ≈ 0.8858

Hard · stretch and reason

7 PROBLEMS

1. Term in x^5 of $(2x - 3)^8$

   ANSWER $-48\,384x^5$

2. Term independent of x in $(x^2 + \frac{1}{x})^6$

   ANSWER 15

3. Term independent of x in $(2x - \frac{1}{x^2})^9$

   ANSWER -672

4. Coefficient of x^2 in $(1 + 2x)^5(1 - x)^3$

   ANSWER 7

5. Find n if the coefficient of x^2 in $(1 + x)^n$ is 45

   ANSWER $n = 10$

6. Show the sum of the coefficients of $(2a - b)^n$ is 1

   ANSWER Put $a = b = 1$

7. Estimate 1.005^{10} to four decimal places

   ANSWER ≈ 1.0511

07 Homework

Homework — 6 tasks

Write the general term before hunting for a specific one.

1. Write rows 0 to 6 of Pascal's triangle.
2. Expand $(x + 3)^4$ and $(2x - 1)^5$.
3. Find the term in x^4 in $(3x + 2)^7$.
4. Find the term independent of x in $(x^3 + \frac{2}{x})^8$.

5. Use the first three terms to estimate 1.03^7 and compare with a calculator.
6. Explain in three sentences why the coefficients of $(a+b)^n$ are the combinations $C(n,k)$.

Statistical data and its types

Where data comes from, how it is grouped, and the four pictures that make it readable.

LESSON 65–67 | 3 × 40 MIN | ALGEBRA 11, §4.3 | IGCSE E20.1–20.2

LESSON CLOCK

Kinds of data	18 min
Grouping	26 min
Histograms	30 min
Cumulative frequency	28 min
Practice	30 min
Homework	3 min

LEARNING OBJECTIVES

- Classify data as qualitative or quantitative, discrete or continuous.
- Build a grouped frequency table with sensible classes.
- Draw a histogram with equal and with unequal class widths.
- Draw and read a cumulative frequency curve.

TEXTBOOK

National	pp. 303–320
Cambridge	Core & Extended, pp. 380–396
Workbook	IGCSE Exercise 20.1

01 Explanation

Kinds of data

KIND	MEANING	EXAMPLE
qualitative	a category, not a number	favourite subject
quantitative discrete	counted; only certain values	number of children
quantitative continuous	measured; any value in a range	height, time

POPULATION AND SAMPLE

The **population** is everyone you want to know about; the **sample** is who you actually measure. A sample is useful only if it is **representative** — asking the football team about sport is biased, however many you ask.

Sample size reduces random error; it does nothing at all about bias. That distinction is the single most useful idea in the whole of statistics.

Grouping

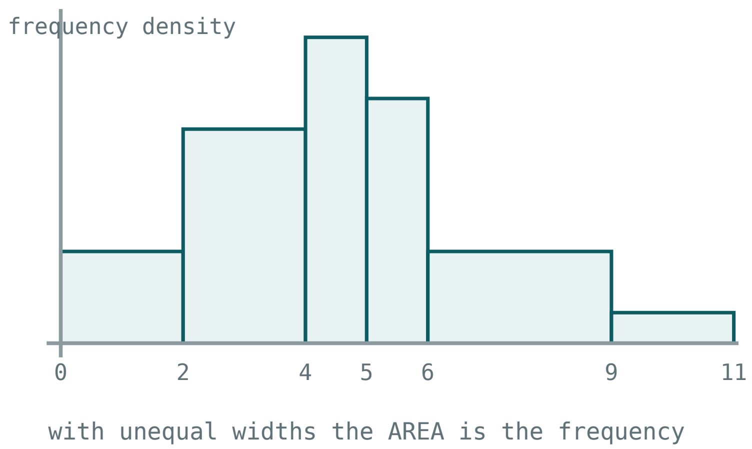
Continuous data must be grouped into **classes**. Three rules make a table readable:

1. Between 5 and 12 classes — fewer hides the shape, more shows only noise.
2. Classes must not overlap and must leave no gaps: $10 \le x < 20$, $20 \le x < 30$.
3. Equal widths where possible; unequal only when the data demands it.

GROUPING LOSES INFORMATION

Once heights are recorded as “*160–170* cm”, the exact values are gone. Every statistic computed from a grouped table is therefore an **estimate**, and should be described as one.

Histograms



With unequal widths the vertical axis is frequency density, and the AREA of each bar is the frequency.

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

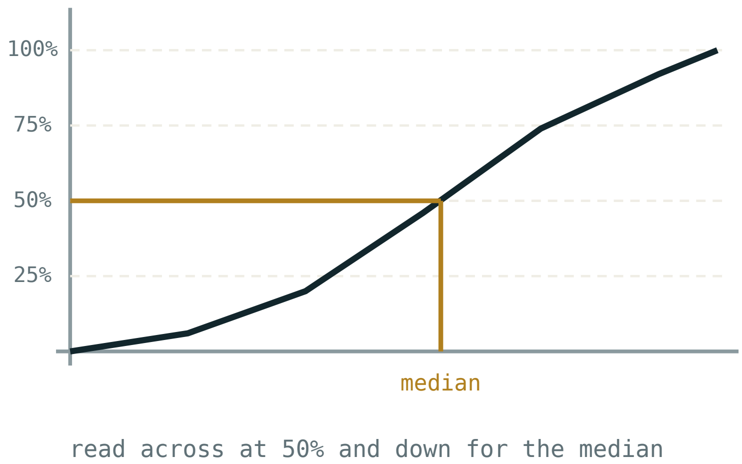
CLASS	FREQUENCY	WIDTH	DENSITY
$0 \le x < 2$	6	2	3
$2 \le x < 4$	14	2	7
$4 \le x < 5$	10	1	10
$5 \le x < 6$	8	1	8
$6 \le x < 9$	9	3	3

A HISTOGRAM IS NOT A BAR CHART

Bars touch, because the scale is continuous. And with unequal widths the **height** is not the frequency — the **area** is. Plotting frequency against unequal widths misrepresents the data, which is exactly what the density is for.

Cumulative frequency

Add the frequencies as you go, and plot each running total against the **upper** end of its class. Join the points with a smooth curve.



Read across at 50% for the median, at 25% and 75% for the quartiles.

READ ACROSS AT	READ DOWN FOR
25%	the lower quartile Q_1
50%	the median
75%	the upper quartile Q_3
a given value	how many are below it

UPPER CLASS BOUNDARIES, ALWAYS

A running total of “fewer than 20” is plotted at 20, not at the midpoint. Plotting at midpoints shifts the whole curve and every quartile with it.

02 Worked examples

EXAMPLE 1 **A class has 30 in $0 \le x < 10$ and 30 in $10 \le x < 40$. Compare the bar heights on a histogram.**

1. Densities $\frac{30}{10} = 3$ and $\frac{30}{30} = 1$.
2. The first bar is three times as tall.
3. But the areas are equal.
   Equal frequencies.

ANSWER

Heights 3 and 1; equal areas

EXAMPLE 2 **Cumulative frequencies $0, 6, 20, 46, 74, 92, 100$ at $0, 2, 4, 6, 8, 10, 11$. Estimate the median.**

1. Total 100; read at 50.
2. Between 46 at 6 and 74 at 8.
3. Interpolate: $6 + \frac{4}{28} \times 2 \approx 6.29$.

ANSWER

≈ 6.3

EXAMPLE 3 **Why is a survey of a school's football team about sport biased?**

1. They were chosen for an interest in sport.
2. They do not represent the school.
3. A bigger football team would not help.

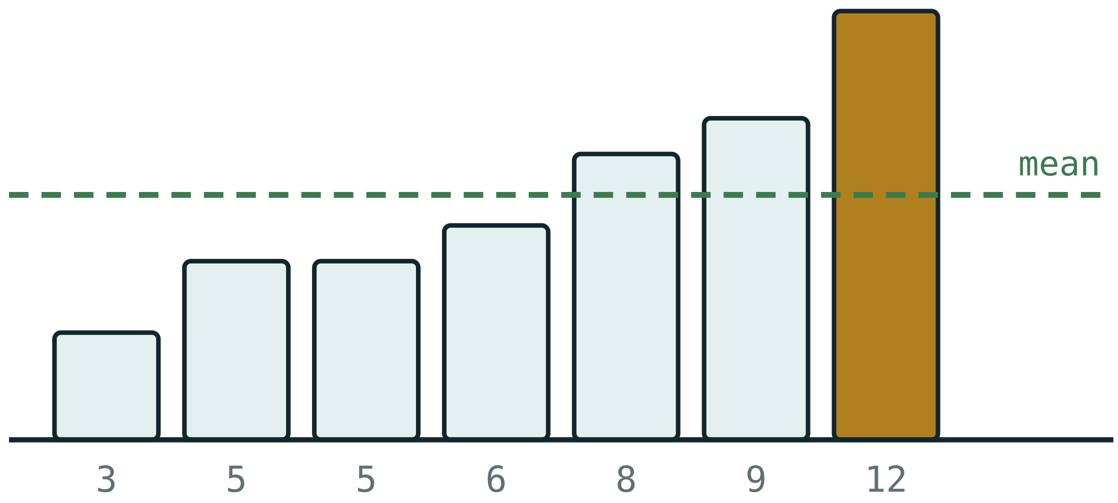
ANSWER

The sample is not representative; size does not fix bias

03 Interactive model

Collect one real set of data from the class and group it live on the board.

Grouping and reading data



values 3, 5, 5, 6, 8, 9, 12 | mean 6.86 | median 6 | mode 5 | range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Population	Bosh to‘plam	Генеральная совокупность
Sample	Tanlanma	Выборка
Qualitative data	Sifat ma’lumot	Качественные данные
Quantitative data	Miqdoriy ma’lumot	Количественные данные
Discrete	Diskret	Дискретные
Continuous	Uzluksiz	Непрерывные
Class interval	Sinf oralig‘i	Классовый интервал
Frequency density	Chastota zichligi	Плотность частоты
Cumulative frequency	To‘plangan chastota	Накопленная частота
Bias in a sample	Tanlanma xatosi	Смещение выборки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Height is:

- qualitative
- discrete
- continuous
- a category

Number of children is:

- qualitative
- discrete
- continuous
- a category

In a histogram with unequal widths the height is:

- frequency
- frequency density
- cumulative frequency
- the class width

Cumulative frequency is plotted at:

- midpoints
- upper class boundaries
- lower boundaries
- anywhere

A bigger sample fixes:

- bias
- random error
- both
- neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is shoe size discrete or continuous?

   ANSWER discrete

2. Is time continuous?

   ANSWER yes

3. Is eye colour quantitative?

   ANSWER no — qualitative

4. Frequency 20, width 4: the density

   ANSWER 5

5. Frequency 15, width 5: the density

   ANSWER 3

6. Density 6, width 3: the frequency

   ANSWER 18

7. Cumulative frequency is plotted at

   ANSWER upper class boundaries

Medium · the standard question

7 PROBLEMS

1. 30 in a width-10 class and 30 in a width-30 class: the heights

   ANSWER 3 and 1

2. Total 100; the median is read at

   ANSWER 50

3. Cumulative $0,6,20,46,74,92,100$: the median

   ANSWER ≈ 6.3

4. Same data: Q_1

   ANSWER ≈ 4.4

5. Same data: Q_3

   ANSWER ≈ 8.1

6. Same data: the interquartile range

   ANSWER ≈ 3.7

7. Why is a football team a biased sample about sport?

   ANSWER Chosen for that very interest

Hard · stretch and reason

7 PROBLEMS

1. A histogram bar has height 4 over $10 \le x < 25$. Its frequency

   ANSWER 60

2. Frequencies $8, 22, 30, 25, 15$ over widths $5, 5, 10, 10, 20$: the densities

   ANSWER $1.6, 4.4, 3, 2.5, 0.75$

3. From a cumulative curve, estimate how many are above 7 if the reading there is 60 of 100

   ANSWER 40

4. Explain why a grouped mean is an estimate

   ANSWER The exact values were lost in grouping

5. Design a sampling method for the heights of a school of 800

   ANSWER Stratified by year group, then random within each

6. Why must histogram bars touch?

   ANSWER The scale is continuous — no gaps in the data

7. A class of width 1 has density 12. Add a class of width 4 and frequency 12: the new density

   ANSWER 3

07 Homework

Homework — 5 tasks

Every histogram must have frequency density on the vertical axis when widths differ.

1. Classify five kinds of data you could collect in your school as qualitative, discrete or continuous.
2. Group these times (seconds) into five classes and draw the histogram: $12, 15, 18, 19, 22, 23, 24, 25, 27, 28, 30, 31, 33, 36, 40$.
3. Frequencies $10, 24, 30, 20, 16$ over widths $5, 5, 10, 10, 20$. Find the frequency densities and draw the histogram.

4. Draw the cumulative frequency curve for the data of task 3 and estimate the median and the quartiles.
5. Explain in three sentences why a larger sample does not correct bias.

Mean, median and mode; deviation and standard deviation

One number for the centre, one for the spread — and knowing which pair to use.

LESSON 68–70 | 3 × 40 MIN | ALGEBRA 11, §4.4 | IGCSE E20.X

LESSON CLOCK

20' | 26' | 28' | 28' | 30' | 8'

Three averages	20 min
Grouped data	26 min
Measures of spread	28 min
Standard deviation	28 min
Practice	30 min
Homework	8 min

LEARNING OBJECTIVES

- Compute the mean, median and mode of raw and of grouped data.
- Choose the appropriate average for a given distribution.
- Compute the range, the interquartile range and the standard deviation.
- Interpret the standard deviation as a typical distance from the mean.

TEXTBOOK

National	pp. 321–338
Cambridge	Core & Extended, pp. 397–412
Workbook	IGCSE Exercise 20.2

01 Explanation

Three averages, three uses

AVERAGE	DEFINITION	BEST WHEN	WEAKNESS
mean	$\frac{\Sigma x}{n}$	data is roughly symmetric	dragged by outliers
median	the middle value	data is skewed or has outliers	ignores most of the data
mode	the commonest value	data is categorical	may not exist or be unique

THE STANDARD ILLUSTRATION

Nine workers earn *4* million so‘m and the director earns *60*. The mean is *9.6* million — a figure nobody earns. The median is *4*. For a question about a typical wage, the median is the honest answer.

Grouped data

The exact values are gone, so each class is represented by its **midpoint**:

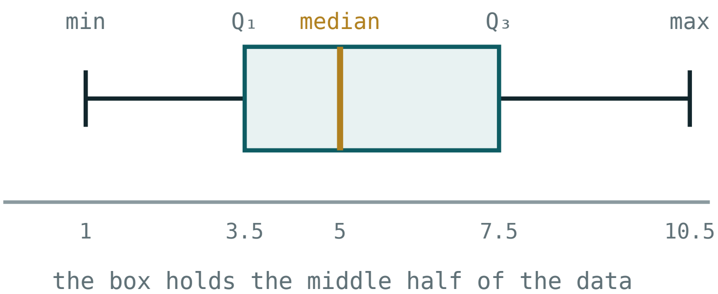
$$\text{estimated mean} = \frac{\Sigma fx}{\Sigma f} \text{ with } x \text{ the class midpoint}$$

CLASS	F	MIDPOINT X	FX
0–10	*4*	*5*	*20*
10–20	*9*	*15*	*135*
20–30	*12*	*25*	*300*
30–40	*5*	*35*	*175*
total	*30*		*630*

Estimated mean $\frac{630}{30} = 21$. The **modal class** is *20–30*; the median lies in the class containing the *15*th value, also *20–30*.

Measures of spread

MEASURE	DEFINITION	RESISTANT TO OUTLIERS?
range	$max - min$	no — it is defined by them
interquartile range	$Q_3 - Q_1$	yes
standard deviation	see below	no



The five-figure summary. The box holds the middle half; its width is the interquartile range.

A **box plot** shows all five numbers at once, and makes two distributions comparable at a glance. A value more than $1.5 \times IQR$ beyond a quartile is conventionally called an **outlier** and marked separately.

Standard deviation

$$\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n} = \frac{\Sigma x^2}{n} - \bar{x}^2 \quad \sigma = \sqrt{\sigma^2}$$

The **variance** σ^2 is the mean of the squared deviations; the **standard deviation** σ is its square root, back in the original units.

WHY SQUARE, AND WHY THEN TAKE THE ROOT

The plain deviations always sum to zero, so they are useless. Squaring removes the signs; the square root brings the answer back to the units of the data, so that "$\sigma = 4$ cm" is meaningful.

Example. $2, 4, 4, 4, 5, 5, 7, 9$: $\bar{x} = 5$, $\Sigma x^2 = 232$, $\sigma^2 = \frac{232}{8} - 25 = 4$, so $\sigma = 2$.

For grouped data the same formula uses f and midpoints: $\sigma^2 = \frac{\Sigma fx^2}{\Sigma f} - \bar{x}^2$.

THE SECOND FORM IS THE ONE TO USE

$\frac{\Sigma x^2}{n} - \bar{x}^2$ needs one pass through the data; the definition needs two, because the mean must be known first. Both give the same number.

02 Worked examples

EXAMPLE 1 **Find the mean, median, mode and standard deviation of *2, 4, 4, 4, 5, 5, 7, 9*.**

1. $\bar{x} = \dfrac{40}{8} = 5$
2. Median $\dfrac{4 + 5}{2} = 4.5$; mode 4.
3. $\Sigma x^2 = 4+16+16+16+25+25+49+81 = 232$
4. $\sigma^2 = 29 - 25 = 4 \Rightarrow \sigma = 2$

ANSWER

$\bar{x} = 5$, median 4.5, mode 4, $\sigma = 2$

EXAMPLE 2 **Estimate the mean of the grouped table with midpoints *5, 15, 25, 35* and frequencies *4, 9, 12, 5*.**

1. $\Sigma f = 30$
2. $\Sigma fx = 20 + 135 + 300 + 175 = 630$
3. $\bar{x} = 21$

ANSWER

21 — an estimate

EXAMPLE 3 **Nine workers earn 4 million and the director 60. Which average should a union quote?**

1. Mean $\frac{96}{10} = 9.6$.

   Nobody earns that.

2. Median 4.

   Nine of ten earn exactly that.

3. The distribution has an extreme outlier.

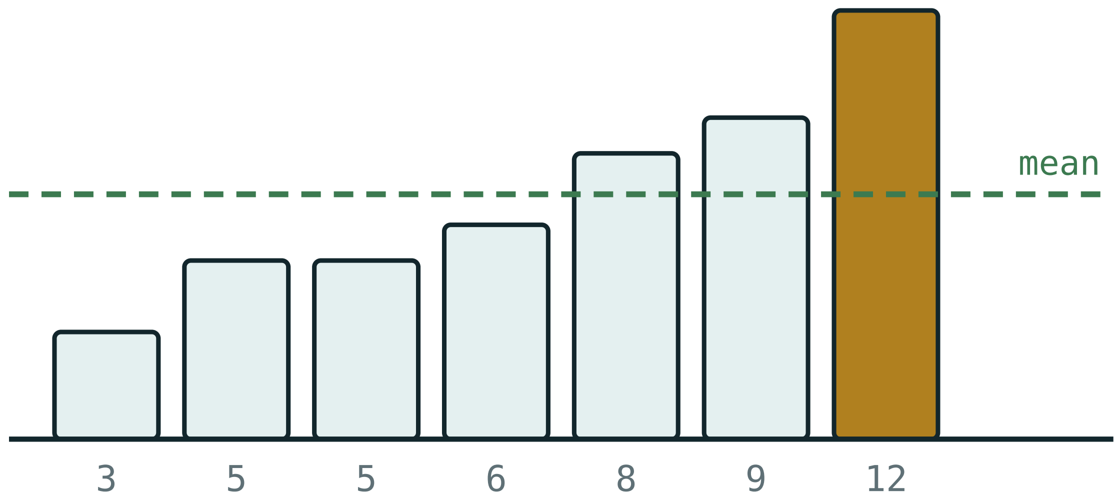
ANSWER

The median — the mean is distorted by one value

03 Interactive model

Compute the mean of the class’s heights, then add one impossible value and recompute.

Centre and spread



values **3, 5, 5, 6, 8, 9, 12** | mean **6.86** | median **6** | mode **5** | range **9**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Mean	Oʻrta arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Modal class	Modal sinf	Модальный класс
Range	Kenglik	Размах
Interquartile range	Kvartillararo kenglik	Межквартильный размах
Deviation	Chetlanish	Отклонение
Variance	Dispersiya	Дисперсия
Standard deviation	Standart chetlanish	Стандартное отклонение
Outlier	Chetlashgan qiymat	Выброс

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which average is dragged by an outlier?

- the mean
- the median
- the mode
- none

For skewed data the best average is:

- the mean
- the median
- the mode
- the range

σ^2 equals:

- $\frac{\Sigma x}{n}$
- $\frac{\Sigma x^2}{n} - \bar{x}^2$
- $\frac{\Sigma(x - \bar{x})}{n}$
- $\sqrt{\Sigma x^2}$

Why are deviations squared?

- tradition
- they otherwise sum to zero
- to make them bigger
- no reason

The interquartile range is:

- $max - min$
- $Q_3 - Q_1$
- 2σ
- the median

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Mean of $3, 5, 7$

   ANSWER 5

2. Median of $3, 5, 7, 9$

   ANSWER 6

3. Mode of $2, 3, 3, 5$

   ANSWER 3

4. Range of $4, 9, 12$

   ANSWER 8

5. Mean of $2, 4, 4, 4, 5, 5, 7, 9$

   ANSWER 5

6. Median of that list

   ANSWER 4.5

7. Mode of that list

   ANSWER 4

Medium · the standard question

7 PROBLEMS

1. σ of $2, 4, 4, 4, 5, 5, 7, 9$

   ANSWER 2

2. Estimated mean from midpoints $5,15,25,35$, frequencies $4,9,12,5$

   ANSWER 21

3. The modal class there

   ANSWER $20–30$

4. σ of $1, 2, 3, 4, 5$

   ANSWER $\sqrt{2} \approx 1.41$

5. Nine at 4 and one at 60: the mean

   ANSWER 9.6

6. Same: the median

   ANSWER 4

7. $Q_3 - Q_1$ for $2,4,5,7,9,11,15$

   ANSWER 7

Hard · stretch and reason

7 PROBLEMS

1. Grouped: midpoints $5,15,25,35$, $f = 4,9,12,5$. Find σ

   ANSWER ≈ 8.6

2. A set of 10 has mean 12 and $\Sigma x^2 = 1560$. Find σ

   ANSWER $\sqrt{12} \approx 3.46$

3. Adding 5 to every value changes the mean and σ how?

   ANSWER Mean +5, σ unchanged

4. Multiplying every value by 3 changes them how?

   ANSWER Both are tripled

5. A set has $\bar{x} = 20$, $\sigma = 4$. What are they after $y = 2x - 3$?

   ANSWER 37 and 8

6. Is 25 an outlier for $Q_1 = 8$, $Q_3 = 14$?

   ANSWER yes — beyond $14 + 9 = 23$

7. Explain why the median ignores most of the data but is still useful

   ANSWER Its resistance is exactly that indifference

07 Homework

Homework — 6 tasks

Say which average you would quote, and why, on every real-context question.

1. Find the mean, median, mode, range and standard deviation of $3, 5, 5, 6, 8, 8, 8, 9, 12, 16$.
2. Estimate the mean and standard deviation of grouped data with midpoints $2, 6, 10, 14$ and frequencies $5, 12, 9, 4$.
3. Draw a box plot for $4, 7, 9, 11, 12, 15, 18, 22, 30$ and identify any outlier.
4. A firm has nine staff on 5 million and a director on 80. Compute both averages and say which is honest.

5. A set has $\bar{x} = 30$ and $\sigma = 6$. Find both after the transformation $y = 3x + 10$.
6. Explain in three sentences why the deviations are squared before averaging.

Studying the relationship between two sets of data

Scatter diagrams, correlation, the line of best fit — and the one warning that must accompany all of it.

LESSON 71–73 | 3 × 40 MIN | ALGEBRA 11, §4.5 | IGCSE E16.1

LESSON CLOCK

18' | 26' | 30' | 28' | 30' | 3

Scatter diagrams	18 min
Describing correlation	26 min
The line of best fit	30 min
Correlation is not causation	28 min
Practice	30 min
Homework	3 min

LEARNING OBJECTIVES

- Plot and read a scatter diagram.
- Describe correlation by type and strength.
- Draw a line of best fit through the mean point and use it to predict.
- Distinguish correlation from causation, and interpolation from extrapolation.

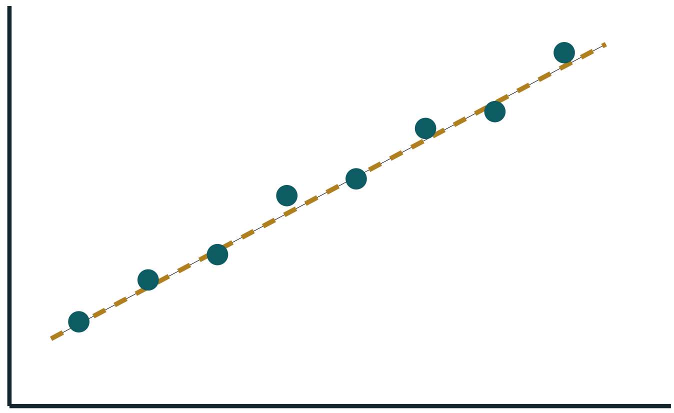
TEXTBOOK

National	pp. 339–356
Cambridge	Core & Extended, pp. 413–424
Workbook	IGCSE Exercise 16.1

01 Explanation

Scatter diagrams

Each individual contributes **one point** (x, y) — the two measurements made on the same person or object. The picture then shows whether the two vary together.



positive correlation

Each point is one individual, measured twice. The pattern, not the points, is the message.

CHOOSE THE AXES DELIBERATELY

If one variable might explain the other, put the explanatory variable on the horizontal axis. “Hours revised” goes on x; “mark” goes on y. It changes nothing mathematically and everything about readability.

Describing correlation

PATTERN	NAME	EXAMPLE
rising, tight	strong positive	height and arm span
rising, loose	weak positive	revision hours and mark
falling, tight	strong negative	age of a car and its price
no pattern	no correlation	shoe size and mathematics mark
a curve	a relationship, but not linear	height of a thrown ball against time

ALWAYS GIVE BOTH WORDS

“Correlation” alone is not an answer. State the **type** — positive or negative — and the **strength** — strong, moderate or weak. Two words, two marks.

The line of best fit

TWO RULES FOR DRAWING IT BY EYE

1 It must pass through the **mean point** $(\bar{x}, \bar{y})$. **2** It should have roughly as many points above it as below, and follow the trend.

Once drawn, it predicts: read up from an x to the line and across to a y. Its gradient is the rate of change — "each extra hour of revision is worth about 4 marks".

PREDICTION	NAME	RELIABLE?
inside the range of the data	interpolation	yes, if the correlation is strong
outside the range	extrapolation	no — the pattern may not continue

Predicting a mark for 3 hours from data covering 1 to 8 hours is interpolation. Predicting for 40 hours is extrapolation, and the line would eventually promise a mark above 100.

Correlation is not causation

THE ONE WARNING THAT MUST ALWAYS ACCOMPANY A CORRELATION

Two variables can move together for three different reasons: one causes the other; both are caused by a third; or it is coincidence. A scatter diagram cannot tell which.

CORRELATED	REAL EXPLANATION
ice-cream sales and drownings	both rise with hot weather
shoe size and reading age in children	both rise with age
number of firefighters and damage	both rise with the size of the fire

Establishing causation needs an **experiment** — changing one variable deliberately and holding others fixed — not a larger data set.

02 Worked examples

EXAMPLE 1 **Describe the correlation between the age of a car and its price.**

1. As age rises, price falls.
   Negative.
2. The points lie fairly close to a line.
   Strong.

ANSWER

Strong negative correlation

EXAMPLE 2 **Data on revision hours $1–8$ gives a best-fit line $y = 42 + 4.5x$. Predict the mark for 5 hours, and for 30.**

1. $42 + 22.5 = 64.5$
   Interpolation — inside the range.
2. $42 + 135 = 177$
   Impossible.
3. The second is extrapolation far outside the data.

ANSWER

≈ 65; the second prediction is invalid

EXAMPLE 3 **Ice-cream sales and drownings are strongly correlated. Does ice cream cause drowning?**

1. Both rise in hot weather.
2. A third variable explains both.
3. Correlation is not causation.

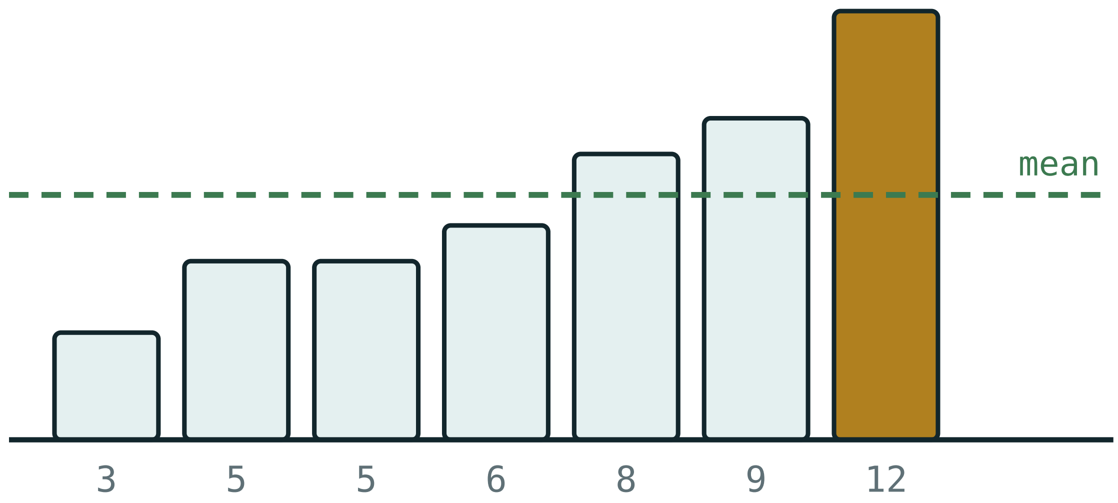
ANSWER

No — temperature is the common cause

03 Interactive model

Plot the class's height against arm span and draw the line through the mean point.

A scatter and its line



values **3, 5, 5, 6, 8, 9, 12** | mean **6.86** | median **6** | mode **5** | range **9**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Bivariate data	Ikki o‘zgaruvchili ma’lumot	Двумерные данные
Scatter diagram	Korrelatsion diagramma	Диаграмма рассеяния
Correlation	Korrelyatsiya	Корреляция
Positive correlation	Musbat korrelyatsiya	Положительная корреляция
Negative correlation	Manfiy korrelyatsiya	Отрицательная корреляция
Line of best fit	Eng mos chiziq	Линия наилучшего соответствия
Mean point	O‘rta nuqta	Средняя точка
Interpolation	Interpolyatsiya	Интерполяция
Extrapolation	Ekstrapolyatsiya	Экстраполяция
Causation	Sababiy bog‘lanish	Причинная связь

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Age of a car against price shows:

- strong positive
- strong negative
- no correlation
- a curve

The line of best fit must pass through:

- the origin
- the mean point
- the first point
- the highest point

Predicting outside the data range is:

- interpolation
- extrapolation
- correlation
- causation

Correlation implies causation:

- always
- never
- sometimes, but it cannot be told from the data
- only if strong

Describing a correlation needs:

- one word
- type and strength
- a number only
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Height and arm span

   ANSWER strong positive

2. Age of a car and its price

   ANSWER strong negative

3. Shoe size and mathematics mark

   ANSWER no correlation

4. Temperature and ice-cream sales

   ANSWER positive

5. The line of best fit passes through

   ANSWER the mean point

6. Predicting inside the range is called

   ANSWER interpolation

7. Predicting outside is called

   ANSWER extrapolation

Medium · the standard question

7 PROBLEMS

1. $y = 42 + 4.5x$; predict for $x = 5$

   ANSWER 64.5

2. Why is the prediction for $x = 30$ invalid?

   ANSWER Far outside the data; it exceeds 100

3. Interpret the gradient 4.5

   ANSWER about 4.5 marks per extra hour

4. Interpret the intercept 42

   ANSWER the predicted mark with no revision

5. Mean point of $(1,3),(2,5),(3,7)$

   ANSWER $(2, 5)$

6. Does a strong correlation prove causation?

   ANSWER no

7. Firefighters and damage: the real cause

   ANSWER the size of the fire

Hard · stretch and reason

7 PROBLEMS

1. Points $(1,4),(2,7),(3,9),(4,12),(5,14)$: the mean point

   ANSWER $(3, 9.2)$

2. Estimate the gradient of the best-fit line for those points

   ANSWER ≈ 2.5

3. Hence estimate y at $x = 3.5$

   ANSWER ≈ 10.5

4. Why must the line pass through the mean point?

   ANSWER The least-squares line always does

5. Give a pair of variables with a strong correlation and no causal link

   ANSWER e.g. stork numbers and birth rates

6. A curve, not a line, fits the data. What does that show?

   ANSWER A relationship, but not a linear one

7. Design an experiment to test whether revision causes better marks

   ANSWER Assign revision time at random and compare groups

07 Homework

Homework — 5 tasks

Every correlation described with two words; every prediction labelled interpolation or extrapolation.

1. Plot the eight points $(1,5),(2,6),(3,9),(4,10),(5,12),(6,14),(7,15),(8,18)$ and describe the correlation.
2. Find the mean point and draw the line of best fit through it.
3. Use your line to predict y at $x = 4.5$ and at $x = 20$, saying which prediction is reliable.
4. Give two pairs of variables that are correlated without one causing the other, with the real explanation.

5. Explain in three sentences the difference between interpolation and extrapolation.

Integration by partial fractions and by parts

Cambridge insert: two techniques that turn an impossible integral into two easy ones.

LESSON 74–76 | 3 × 40 MIN | ALGEBRA 11, §3.2 (EXTENSION) | P2 · 8.1–8.3

LESSON CLOCK

20' | 30' | 30' | 30' | 26' | 4

Why split a fraction	20 min
Partial fractions	30 min
Integrating them	30 min
By parts	30 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Split a proper rational function into partial fractions.
- Integrate using partial fractions.
- State and apply the integration-by-parts formula.
- Choose which factor to differentiate and which to integrate.

TEXTBOOK

National	pp. 357–364
Cambridge	Pure Mathematics 2 & 3, pp. 160–178
Workbook	P2 Exercise 8A–8C

01 Explanation

Why split a fraction

$\int \frac{5x-1}{x^2-1}\,dx$ has no obvious antiderivative. But the integrand can be written as a sum of two much simpler ones:

$$\frac{5x-1}{x^2-1} = \frac{2}{x-1} + \frac{3}{x+1}$$

and each of those integrates to a logarithm at sight.

THE ONE STANDARD INTEGRAL YOU NEED

$$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\,ln|ax+b| + C$$

Every partial-fraction integral reduces to copies of this.

Splitting the fraction

The fraction must be **proper** — numerator of lower degree than the denominator. If it is not, divide first.

DENOMINATOR	FORM OF THE SPLIT
$(x-a)(x-b)$	$\frac{A}{x-a} + \frac{B}{x-b}$
$(x-a)^2$	$\frac{A}{x-a} + \frac{B}{(x-a)^2}$
$(x-a)(x^2+c)$	$\frac{A}{x-a} + \frac{Bx+C}{x^2+c}$

The cover-up method. To find A in $\frac{5x-1}{(x-1)(x+1)}$, cover the factor $(x-1)$ and put $x = 1$ into what remains: $\frac{5(1)-1}{1+1} = 2$. Similarly $B = \frac{5(-1)-1}{-1-1} = 3$. Two substitutions, no simultaneous equations.

Integrating them

$$\int \frac{5x-1}{x^2-1}\,dx = \int\left(\frac{2}{x-1} + \frac{3}{x+1}\right)dx$$

$$= 2\ln|x-1| + 3\ln|x+1| + C$$

THE MODULUS SIGNS MATTER

$\ln$ is defined only for positive arguments; $\ln|x-1|$ covers both sides of $x = 1$. Dropping the bars loses a mark and, in a definite integral, can lose the answer.

A repeated factor gives a term that is **not** a logarithm: $\int \frac{1}{(x-2)^2}\,dx = -\frac{1}{x-2} + C$ — the power rule, not the logarithm.

Integration by parts

$$\int u\,\frac{dv}{dx}\,dx = uv - \int v\,\frac{du}{dx}\,dx$$

It is the product rule integrated. Its purpose is to **replace one integral by an easier one** — so the choice of u decides whether it helps.

CHOOSING U

Take u to be the factor that gets **simpler when differentiated**: a power of x before an exponential; a logarithm before anything.

Example. $\int x\, e\^x\, dx$: take $u = x$, $\frac{dv}{dx} = e\^x$. Then $\int x\, e\^x\, dx = x\, e\^x - \int e\^x\, dx = e\^x(x - 1) + C$.

Example. $\int \ln x\, dx$: take $u = \ln x$, $\frac{dv}{dx} = 1$. Then $= x \ln x - \int 1\, dx = x \ln x - x + C$ — an integral with only one factor, done by parts.

A BAD CHOICE MAKES IT WORSE

With $u = e\^x$ in the first example, the new integral has x^2 in it. If the second integral is harder than the first, swap the choice and start again.

02 Worked examples

EXAMPLE 1 **Split $\frac{5x - 1}{x^2 - 1}$ into partial fractions.**

1. $x^2 - 1 = (x-1)(x+1)$
2. Cover $(x-1)$, put $x = 1$: $\frac{4}{2} = 2$.
3. Cover $(x+1)$, put $x = -1$: $\frac{-6}{-2} = 3$.

ANSWER

$$\frac{2}{x - 1} + \frac{3}{x + 1}$$

EXAMPLE 2 **Find $\int \frac{5x - 1}{x^2 - 1}\,dx$.**

1. Use the split above.
2. $\int \frac{2}{x-1}\,dx = 2\ln|x - 1|$
3. $\int \frac{3}{x+1}\,dx = 3\ln|x + 1|$

ANSWER

$2\ln|x - 1| + 3\ln|x + 1| + C$

EXAMPLE 3 **Find $\int x \cos x\,dx$.**

1. $u = x,\ \frac{dv}{dx} = \cos x.$

   x simplifies when differentiated.
2. $v = \sin x,\ \frac{du}{dx} = 1.$
3. $x \sin x - \int \sin x\,dx$
4. $= x \sin x + \cos x + C$

ANSWER

$x \sin x + \cos x + C$

03 Interactive model

Do the cover-up method on the board twice — it is faster than any alternative.

This model could not start: Cannot read properties of undefined (reading 'start')

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Partial fractions	Elementar kasrlar	Простейшие дроби
Proper fraction	To‘g‘ri kasr	Правильная дробь
Improper fraction	Noto‘g‘ri kasr	Неправильная дробь
Linear factor	Chiziqli ko‘paytuvchi	Линейный множитель
Repeated factor	Takrorlanuvchi ko‘paytuvchi	Кратный множитель
Integration by parts	Bo‘laklab integrallash	Интегрирование по частям
Cover-up method	Yopish usuli	Метод вычёркивания
Reduction	Kamaytirish	Понижение
Natural logarithm	Natural logarifm	Натуральный логарифм

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\int \frac{1}{x-3}\,dx$ is:

- $\ln(x-3)+C$
- $\ln|x-3|+C$
- $-\frac{1}{(x-3)^2}+C$
- $\frac{1}{2}(x-3)^2+C$

$\frac{1}{(x-1)(x+2)}$ splits into:

- $\frac{A}{x-1}+\frac{B}{x+2}$
- $\frac{Ax+B}{(x-1)(x+2)}$
- $\frac{A}{(x-1)(x+2)}$
- it does not split

Integration by parts is:

- the chain rule reversed
- the product rule reversed
- the quotient rule reversed
- a new rule

For $\int x\, e^{\wedge}x\, dx$ take u to be:

- $e^{\wedge}x$
- x
- $x\, e^{\wedge}x$
- either

$\int \ln x\, dx$ equals:

- $\frac{1}{x} + C$
- $x \ln x - x + C$
- $\ln x + C$
- $\frac{(\ln x)^2}{2} + C$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\int \frac{1}{x - 2}\, dx$

ANSWER $\ln|x - 2| + C$

2. $\int \frac{1}{2x + 1}\, dx$

ANSWER $\frac{1}{2} \ln|2x + 1| + C$

3. $\int \frac{3}{x}\, dx$

ANSWER $3 \ln|x| + C$

4. $\int \frac{1}{(x-1)^2}\, dx$

ANSWER $-\frac{1}{x - 1} + C$

5. Split $\frac{1}{x(x+1)}$

ANSWER $\frac{1}{x} - \frac{1}{x+1}$

6. $\int x\, e^\wedge x\, dx$

ANSWER $e^\wedge x(x - 1) + C$

7. $\int \ln x\, dx$

ANSWER $x \ln x - x + C$

Medium · the standard question

7 PROBLEMS

1. Split $\dfrac{5x - 1}{x^2 - 1}$

ANSWER $\dfrac{2}{x-1} + \dfrac{3}{x+1}$

2. $\displaystyle\int \frac{5x - 1}{x^2 - 1}\, dx$

ANSWER $2 \ln|x-1| + 3 \ln|x+1| + C$

3. Split $\dfrac{3x + 5}{(x+1)(x+3)}$

ANSWER $\dfrac{1}{x+1} + \dfrac{2}{x+3}$

4. $\int x \cos x\, dx$

ANSWER $x \sin x + \cos x + C$

5. $\int x \sin x\, dx$

ANSWER $-x \cos x + \sin x + C$

6. $\int x^2 e\text{^}x\, dx$

ANSWER $e\text{^}x(x^2 - 2x + 2) + C$

7. Split $\dfrac{4}{x^2 - 4}$

ANSWER $\dfrac{1}{x-2} - \dfrac{1}{x+2}$

Hard · stretch and reason

7 PROBLEMS

1. Split $\dfrac{x + 7}{(x-1)(x+2)^2}$

ANSWER $\dfrac{\frac{8}{9}}{x-1} - \dfrac{\frac{8}{9}}{x+2} - \dfrac{\frac{5}{3}}{(x+2)^2}$

2. $\int \dfrac{x^2 + 1}{x^2 - 1}\, dx$

ANSWER $x + \ln|x-1| - \ln|x+1| + C$

3. $\int x \ln x\, dx$

ANSWER $\dfrac{x^2}{2} \ln x - \dfrac{x^2}{4} + C$

4. $\int x^2 \ln x\, dx$

ANSWER $\dfrac{x^3}{3} \ln x - \dfrac{x^3}{9} + C$

5. $\int_2^3 \dfrac{1}{x(x-1)}\, dx$

ANSWER $\ln \dfrac{4}{3}$

6. $\int e^x \sin x\, dx$

ANSWER $\dfrac{e^x}{2} (\sin x - \cos x) + C$

7. Explain why the cover-up method works

ANSWER Multiplying by the factor and substituting its root kills the other terms

07 Homework

Homework — 6 tasks

Modulus signs inside every logarithm; state which factor you chose as u and why.

1. Split $\frac{7x - 1}{x^2 - 1}$ and $\frac{2x + 5}{(x+1)(x+4)}$ into partial fractions.
2. Find $\int \frac{7x - 1}{x^2 - 1}\, dx$.
3. Find $\int \frac{1}{(x-3)^2}\, dx$ and explain why it is not a logarithm.
4. Find $\int x\, e^{(2x)}\, dx$.
5. Find $\int x^2 \ln x\, dx$.
6. Explain in three sentences how you decide which factor to call u.

Control work 6, and the quarter review

Combinatorics, the binomial theorem and statistics in one paper, then the map of the whole quarter.

LESSON 77–78 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 6 | IGCSE E20 REVIEW

LESSON CLOCK

3 | 42' | 10' | 20' | 10' | 5'

Instructions	3 min
The paper	42 min
Answers	10 min
Rewrite	20 min
Concept map	10 min
Targets	5 min

LEARNING OBJECTIVES

- Apply the combinatorial and statistical methods under time.
- Decide whether order matters, unprompted.
- Build a concept map of the quarter.
- Set a personal target for Quarter IV.

TEXTBOOK

National	pp. 365–368
Cambridge	Core & Extended, pp. 425–428
Workbook	Control paper F

01 Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	From 9 people: a team of 4; then a captain, a vice-captain and two others	*5*	L60–62
2	How many arrangements has SAMARQAND? And with the two A's together?	*5*	L60–62
3	Find the term in x^4 in $(2x - 1)^7$	*4*	L63–64
4	Estimate the mean and standard deviation of a grouped table	*6*	L68–70
5	Draw a cumulative frequency curve and read off the median and quartiles	*6*	L65–67
6	Describe a given correlation and say why a prediction at $x = 40$ is invalid	*4*	L71–73

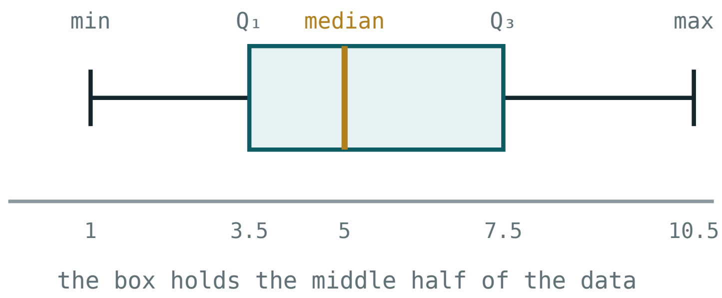
WHERE THE MARKS HIDE

Q1 gives one mark for saying whether order matters in each part. Q6 gives two words — type and strength — and two for the extrapolation reason. Five of thirty are for words, not numbers.

The concept map

Six boxes, links as sentences:

- **antiderivative** → **definite integral** — "Newton–Leibniz: $F(b) - F(a)$"
- **definite integral** → **area and volume** — "upper minus lower; $\pi\int y^2\,dx$"
- **no antiderivative** → **trapezium rule** — "straight chords, error $\propto h^2$"
- **counting rules** → $\boldsymbol{C(n,k)}$ — "order matters, or it does not"
- $\boldsymbol{C(n,k)}$ → **binomial theorem** — "Pascal's triangle is the coefficients"
- **data** → **centre and spread** — "mean and σ, or median and IQR"



The picture behind the last box.

Looking forward

Quarter IV is probability and distributions, then the Cambridge revision blocks on complex numbers and differential equations. The combinatorics of this quarter is what makes probability computable, and the integration is what solves the differential equations.

> ONE HABIT TO CARRY FORWARD
>
> Ask “does order matter?” before every counting question, and “is this the mean or the median’s job?” before every statistic. Both are one-sentence decisions that determine the whole answer.

02 Worked examples

EXAMPLE 1 **Model answer, Q1: from 9 people, a team of 4; then a captain, vice-captain and two others.**

1. A team: order does not matter.
   $C(9,4) = 126$
2. With roles: choose the captain, then the vice, then two others.
   $9 \times 8 \times C(7,2)$
3. $= 9 \times 8 \times 21 = 1512$

> ANSWER
>
> 126; 1512

EXAMPLE 2 **Model answer, Q2: arrangements of SAMARQAND.**

1. 9 letters with A×3.

$$\frac{9!}{3!} = 60\ 480$$

2. With the three A's together: treat them as one block.

$$7! = 5040$$

3. (With only two A's together the count differs.)

ANSWER

60 480; *5040* with all three A's together

EXAMPLE 3 **Model answer, Q3: the term in x^4 in $(2x - 1)^7$.**

1. $7 - k = 4 \Rightarrow k = 3$
2. $C(7,3) = 35$
3. $35\,(2x)^4\,(-1)^3$
4. $= 35 \times 16 \times (-1)\, x^4 = -560x^4$

ANSWER

$-560x^4$

03 Interactive model

Work Q1 and Q3 on the board, stating the order question aloud each time.

The quarter in ten questions

$\int_0^3 x^2\,dx$:

- 9
- 27
- 3
- 6

Area between $y = 4x$ and $y = x^2$:

- $\frac{16}{3}$
- $\frac{32}{3}$
- 16
- 8

Volume from $y = \sqrt{x}$ on $[0, 4]$:

- 4π
- 8π
- 16π
- 2π

4 strips need how many ordinates?

- 3
- 4
- 5
- 8

A team of 4 from 9:

- 126
- 3024
- 36
- 24

Arrangements of SAMARQAND:

- $9!$
- $\frac{9!}{3!}$
- $\frac{9!}{2!}$
- $7!$

Term in x^4 of $(2x - 1)^7$:

- $560x^4$
- $-560x^4$
- $35x^4$
- $-35x^4$

Which average resists outliers?

- the mean
- the median
- the mode
- none

σ^2 equals:

- $\frac{\Sigma x}{n}$
- $\frac{\Sigma x^2}{n} - \bar{x}^2$
- $\sqrt{\Sigma x^2}$
- $\frac{\Sigma(x - \bar{x})}{n}$

Predicting outside the data range is:

- interpolation
- extrapolation
- correlation
- valid

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Permutation	O‘rin almashtirish	Перестановка
Combination	Kombinatsiya	Сочетание
Standard deviation	Standart chetlanish	Стандартное отклонение
Correlation	Korrelyatsiya	Корреляция
Self-assessment	O‘z-o‘zini baholash	Самооценка
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first question in any counting problem is:

- how many?
- does order matter?
- is it a factorial?
- is it large?

A grouped mean is:

- exact
- an estimate
- wrong
- the median

A definite integral’s answer is:

- a family
- a number
- a function
- a constant of integration

Quarter IV begins with:

- integration
- probability
- vectors
- logarithms

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $C(9,4)$

   ANSWER 126

2. $A(9,2)$

   ANSWER 72

3. Arrangements of SAMARQAND

   ANSWER $60\,480$

4. Term in x^4 of $(2x - 1)^7$

   ANSWER $-560x^4$

5. $\int_0^3 x^2\,dx$

   ANSWER 9

6. Mean of $4, 6, 8, 10$

   ANSWER 7

7. Median of $4, 6, 8, 10, 30$

   ANSWER 8

Medium · the standard question

7 PROBLEMS

1. A captain, a vice and two others from 9

ANSWER 1512

2. Arrangements of SAMARQAND with the three A's together

ANSWER 5040

3. Area between $y = 4x$ and $y = x^2$

ANSWER $\frac{32}{3}$

4. Volume from $y = \sqrt{x}$ on $[0, 4]$

ANSWER 8π

5. σ of $2, 4, 4, 4, 5, 5, 7, 9$

ANSWER 2

6. Grouped mean, midpoints $5,15,25,35$, $f = 4,9,12,5$

ANSWER 21

7. Estimate $\int_0^4 x^2\,dx$ with 4 strips

ANSWER 22

Hard · stretch and reason

7 PROBLEMS

1. From 7 men and 5 women, committees of 5 with at least 3 women

   ANSWER *246*

2. Term independent of x in $(2x - \frac{1}{x^2})^9$

   ANSWER -672

3. Grouped σ with midpoints $5,15,25,35$, $f = 4,9,12,5$

   ANSWER ≈ 8.6

4. Area enclosed by $y = x^3$ and $y = x$

   ANSWER $\frac{1}{2}$

5. $\int \frac{5x-1}{x^2-1}\,dx$

   ANSWER $2\ln|x-1| + 3\ln|x+1| + C$

6. $\int x \cos x \, dx$

   ANSWER $x \sin x + \cos x + C$

7. Why is a prediction at $x = 40$ invalid for data on $[1, 8]$?

   ANSWER Extrapolation far outside the range

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter IV.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. From 8 men and 6 women, how many committees of 5 contain at least 2 women? And find the term in x^3 of $(3x - 2)^6$.

4. Write your target for Quarter IV in one checkable sentence, and date it.